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Thermal Transport in 2D Semiconductors—Considerations for Device Applications

Yunshan Zhao, Yongqing Cai, Lifa Zhang, Baowen Li, Gang Zhang, and John T. L. Thong**

The discovery of graphene has stimulated the search for and investigations into other 2D materials because of the rich physics and unusual properties exhibited by many of these layered materials. Transition metal dichalcogenides (TMDs), black phosphorus, and SnSe among many others, have emerged to show highly tunable physical and chemical properties that can be exploited in a whole host of promising applications. Alongside the novel electronic and optical properties of such 2D semiconductors, their thermal transport properties have also attracted substantial attention. Here, a comprehensive review of the unique thermal transport properties of various emerging 2D semiconductors is provided, including TMDs, black- and blue-phosphorene among others, and the different mechanisms underlying their thermal conductivity characteristics. The focus is placed on the phonon-related phenomena and issues encountered in various applications based on 2D semiconductor materials and their heterostructures, including thermoelectric power generation and electron–phonon coupling effect in photoelectric and thermal transistor devices. A thorough understanding of phonon transport physics in 2D semiconductor materials to inform thermal management of next-generation nanoelectronic devices is comprehensively presented along with strategies for controlling heat energy transport and conversion.

1. Introduction

Since the first discovery of monolayer carbon atoms arranged in honeycomb structures,[1,2] 2D graphene has inspired the community of materials scientists and opened up numerous competitive applications in electronics,[3] spintronics,[4] and valleytronics.[5] Due to its strong in-plane covalent bonds formed by sp^2-hybridized carbon arrangement, graphene exhibits many advantages compared with traditional materials, like super high electrical conductivity (high mobility of charge carriers)[6] and superior thermal conductivity.[7,8] As a zero bandgap semimetal,[9,10] graphene is limited for field-effect-transistor (FET) applications since the carrier channel cannot be fully closed by gate voltage, resulting in a low working on/off ratio (<10) that is typical for graphene-channel FETs.[11] Following graphene, there is an increasing abundance of other members such as transition metal dichalcogenides (TMDs), black phosphorus (BP) etc. that have joined the 2D family, which also show excellent transport properties, good mechanical flexibility and most importantly, interesting layer-tunable bandgaps. Recent high-field measurements of MoS_2[12,13] and black phosphorus[14] show better performances than silicon in terms of both critical field and saturation velocity. Therefore, since discovery these kinds of 2D semiconductors are believed to be good candidates for post-silicon semiconductor technology[15–17] due to their atomically thin channel and freedom from surface dangling bonds.

For 2D semiconductors with a honeycomb lattice that are noncentrosymmetric, such as TMDs, valley-contrasting electronic physics has led to the emergence of valleytronics, where the low-energy electronic states with handedness properties have been proven to create chiral optical excitations.[18,19] Recently, it has been theoretically predicted[20] and then experimentally observed[21] that phonons in these materials (WSe_2) can have handedness or chirality as well. Chiral phonons are also observed in graphene/hexagonal boron nitride heterostructure,[22] where chiral phonons are tunable and determine the selection rules of electronic intervalley scattering. This gives rise to potential applications in

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DOI: 10.1002/adfm.201903929

二维半导体中的热传输——器件应用的考虑

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石墨烯的发现激发了人们对其他二维材料的探索，因为这些层状材料具有丰富的物理特性和不寻常的特性。
过渡金属二硫属化物（TMDs）、黑磷和硒化亚锡等材料展现出高度可调的物理和化学特性，这些特性可应用于众多前景广阔的技术领域。除了这类二维半导体材料新颖的电子和光学特性外，其热输运特性也备受关注。本文对包括TMDs、黑磷和蓝磷烯在内的多种新兴二维半导体材料的独特热输运特性进行了全面综述，并探讨了其热导率特征背后的多种机制。重点聚焦于基于二维半导体材料及其异质结构的各种应用中遇到的声子相关现象与问题，包括热电发电以及光电和热晶体管器件中的电子-声子耦合效应。文章还系统阐述了对二维半导体材料中声子输运物理的深入理解，以指导下一代纳米电子器件的热管理，并提出了控制热能传输与转换的策略。

1. 介绍

自单层碳原子以蜂窝状结构排列的首次发现以来，[1,2] 二维石墨烯不仅激发了材料科学界的灵感，更在电子学、[3] 自旋电子学 [4] 和谷电子学领域催生了众多竞争性应用。[5] 由于 ^{2} sp杂化碳原子形成的强平面共价键，石墨烯相较于传统材料展现出诸多优势，例如超高电导率（载流子迁移率高）[6] 和优异的热导率。[7,8] 作为零带隙半金属，[9,10] 石墨烯在场效应晶体管（FET）应用中存在局限——栅极电压无法完全关闭载流子通道，导致其工作开关比（<10）偏低，这是石墨烯沟道场效应晶体管的典型特征。[11] 随着石墨烯的出现，过渡金属二硫属化物（TMDs）、黑磷（BP）等新材料也相继加入二维材料家族，其发展势头日益强劲。

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DOI: 10.1002/adfm.201903929

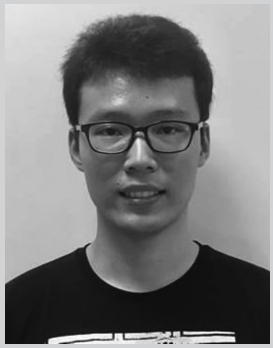
还表现出优异的输运特性、良好的机械柔韧性，最重要的是，具有有趣的层可调带隙。最近对MoS_2 [12，13]和黑磷 [14] 的高场测量表明，它们在临界场和饱和速度方面都优于硅。因此，自发现以来，这类二维半导体因其原子级薄的通道和无表面悬键的特性，被认为是有希望成为后硅半导体技术的候选材料 [15–17] 。

对于具有蜂窝状晶格结构且非中心对称的二维半导体材料（如过渡金属硫属化合物TMDs），谷间对比电子物理现象催生了谷电子学领域。研究表明，具有手性特性的低能电子态能够产生手性光学激发。[18,19] 近期理论预测 [20] 与实验观测 [21] 表明，这类材料（如二硒化钨WSe_2）中的声子同样具有手性特征。在石墨烯/六方氮化硼异质结结构中也观测到手性声子，[22] 其可调谐特性决定了电子谷间散射的选择定则。这一发现为相关技术应用开辟了新途径。

valleytronics and phonon-chirality-based phononics.[23] In quantum dots of WSe_2, the entanglement between chiral phonons and optical excitation have also been reported very recently.[24]

Alongside the novel phononics and electronic/optical properties of 2D semiconductors, their thermal transport properties have drawn considerable interest as well. They are ideal platforms to study fundamental energy carrier transport and provide new directions for thermal energy control and management. As atomically thin layered structures held together by weak van der Waals forces, some 2D semiconducting materials show strong quantum confinement effect[25,26] while the weak in-plane covalent bonds result in much lower thermal conductivity compared with that of graphene,[7] which is beneficial for thermoelectric (TE) energy conversion purposes, where low thermal conductivity is needed to achieve high efficiency.[27–29] On the other hand, with device scaling and increasing integration in modern integrated circuits, the power dissipation density has grown significantly, and high-temperature hot spots can lead to performance degradation as well as long-term reliability issues. Here a material with higher thermal conductivity alleviates the dissipation of the Joule heat originating from the active device, interconnects, and contacts. Typically, for heat conduction, Fourier's law is employed to characterize the heat conduction ability of a solid, and can be expressed as as $J = -\kappa\nabla T$, where J is the heat flux across the system, κ is the thermal conductivity, and ∇T is the temperature gradient. A better understanding of phonon transport in 2D semiconductors would aid in the design of novel devices that exploit such materials.

With increasing interest in new 2D materials, various fabrication methods have been proposed, namely top-down methods like mechanical/liquid exfoliation, and bottom-up methods like chemical vapor deposition (CVD) and wet-chemical synthesis.[30–32] Generally exfoliated specimens are of better quality in terms of crystallinity that is consistent with their bulk counterparts while synthesis approaches yield the quantity needed for industrial applications.[33,34] Regarding to the thermal properties of 2D materials, many research papers and review articles have been published. For theoretical calculation and simulation modeling, the reader can refer to refs. [35,36]. For thermal measurement methods, see refs. [37,38]. For a detailed survey of TE applications of 2D materials, readers can refer to refs. [39,40]. Our review takes a perspective from the various thermally related issues encountered in FETs based on 2D semiconductors, such as thermal management in the active device regions and interfacial thermal resistance across various contacts and interfaces. This article reviews phonon transport phenomena that are relevant not only to FET applications, but also to various other intriguing applications based on 2D semiconductor materials and their heterostructures. Specifically, we will include a discussion of several interesting works about functionalized 2D semiconductor materials for thermal transistor applications that have been reported recently. Our review aims to provide a thorough understanding of thermal transport issues in various applications based on 2D semiconductor materials, to pave the way for better thermal management and heat dissipation when designing next-generation nanoelectronic and energy conversion devices.



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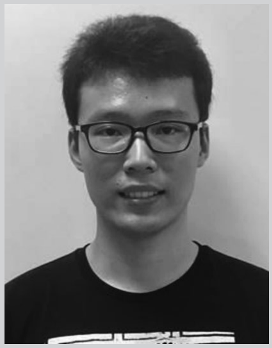
The outline of this review is as follows. A 2D semiconductor-based FET is shown in **Figure 1**, where various elements constituting the 2D semiconductor family are represented, like 1T- and 2H-MoS_2, BP, InSe, SnSe, and so on. Section 2 focuses on the unique thermal transport properties of these 2D semiconductors. As a representative TMDs, MoS_2 shows interesting thermal transport along the in-plane and out-of-plane directions, and the influence of layer-thickness and isotopes on its thermal transport will be discussed. Discussions on the anisotropic thermal transport behavior of BP and the ultralow thermal conductivity of SnSe for TE applications are included in this section. Electrical contacts to 2D semiconductor channels are typically made with

谷电子学和基于声子手性的声子学。[23] 在WSe_2的量子点中，手性声子与光激发之间的纠缠也最近被报道。[24]

二维半导体不仅展现出独特的声子学特性和电子/光学性能，其热输运特性同样备受关注。这类材料堪称研究基本能量载体传输的理想平台，为热能调控与管理开辟了新方向。作为由微弱范德华力维系的原子级薄层结构，部分二维半导体材料展现出显著的量子限制效应[25,26]，而其平面内共价键的弱结合特性导致热导率远低于石墨烯。[7] 这种特性反而有利于热电（TE）能量转换应用——该领域需要低热导率来实现高效能。[27–29] 另一方面，随着现代集成电路器件的微型化和集成度提升，功耗密度显著增加，高温热点不仅会降低器件性能，还可能引发长期可靠性问题。此时，具备更高热导率的材料能有效缓解来自有源器件、互连结构及接触点的焦耳热耗散。在热传导研究中，傅里叶定律是表征固体导热性能的常用理论，其数学表达式为$J=-\kappa\nabla T$。其中，J表示系统内的热流密度，κ为热导率，∇T则是温度梯度。深入理解二维半导体中的声子输运机制，将为开发基于这类材料的新型器件提供重要理论支撑。

随着新型二维材料研究热度持续升温，学界提出了多种制备方法：自上而下的机械/液相剥离法，以及自下而上的化学气相沉积（CVD）和湿化学合成法。[30–32] 通常剥离样品的晶体质量更优，其结晶度与块体材料相当，而合成方法则能提供工业应用所需的量产规模。[33,34] 关于二维材料的热学特性，已有大量研究论文和综述文章发表。理论计算与模拟建模可参考文献[35,36]，热测量方法详见文献[37,38]，二维材料热电应用的详细综述可参阅文献[39,40]。本文综述聚焦二维半导体场效应晶体管（FET）面临的各类热学问题，包括有源器件区域的热管理，以及各类接触界面与接触面之间的界面热阻分析。本文系统梳理了声子输运现象的相关研究，这些现象不仅对场效应晶体管（FET）应用至关重要，更广泛涉及基于二维半导体材料及其异质结构的多种创新应用。特别值得关注的是，我们重点探讨了近期发表的几项关于功能化二维半导体材料在热晶体管领域应用的前沿研究成果。通过系统性综述，本文旨在为基于二维半导体材料的各类应用提供热输运问题的全面认知，为下一代纳米电子器件和能量转换装置的设计优化，特别是在热管理与散热性能提升方面，奠定理论基础。



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二维材料，重点研究其电学与

热学特性，尤其关注电子器件应用。

本综述的结构如下。**图1** 展示了一种基于二维半导体的场效应晶体管，其中展示了构成二维半导体家族的各种元素，如1T-和2H-MoS_2、BP、InSe、硒化亚锡等。第2节重点讨论这些二维半导体独特的热输运特性。作为典型的过渡金属硫属化合物（TMDs），MoS_2展现出沿面内和面外方向的有趣热输运特性，本节将讨论层厚度和同位素对其热输运的影响。此外，本节还讨论了BP的各向异性热输运行为以及硒化亚锡在热电应用中极低的热导率。通常，与二维半导体通道的电接触是通过以下方式实现的

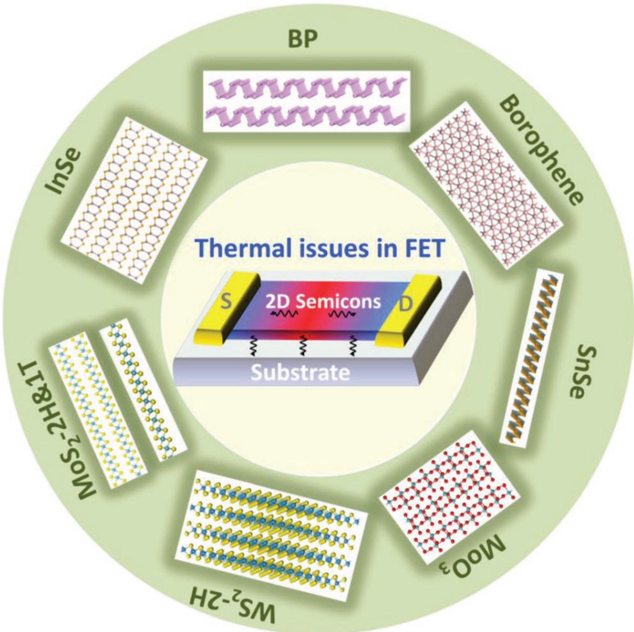

Figure 1. Representative 2D semiconductors that have been applied to FET devices. Various 2D semiconductors are shown, such as 1T- and 2H-MoS_2, BP, InSe, and SnSe.

metals and transport across such interfaces, including MoS_2/contact and MoS_2/substrate will be addressed in Section 3. In addition to these primary considerations in FET thermal management, thermal phenomena related to other device applications of 2D materials, i.e., thermoelectrics, electron–phonon coupling in photoelectric phenomena, effect of electrical field as well as strain on thermal property and thermal functional devices, like thermal transistor, are discussed in Section 4.

2. Thermal Properties in 2D Semiconductors

2.1. Transition Metal Dichalcogenides

Graphene is limited in its electronic device applications due to its lack of an electrical bandgap[1] and this has prompted the search for similar 2D materials with semiconducting properties. Transition metal dichalcogenides having the chemical formula of MX_2, where M denotes a transition metal element and X represents a chalcogen element as shown in **Figure 2**, are promising candidates. With a reduction in the material thickness from bulk to monolayer, TMDs show an indirect- to direct-bandgap transformation, thus enabling potential applications for electronic and optoelectronic devices. While monolayer TMDs are semiconducting with bandgaps around 1–2 eV, the corresponding bulk TMDs show quite diverse properties—from insulators and semiconductors to semimetals and true metals.[31]

Because of the difference in lattice structure of TMDs compared with graphene, TMDs have different phonon transport behaviors, and show much lower thermal conductivity in theory and through experimental measurements. In contrast to dimension-dependence of thermal conductivity in graphene, thermal conductivity of TMDs typically shows near independence on size and roughness due to the very short intrinsic phonon mean free path (MFP).[41] Moreover, compared to the strong sp2 C−C covalent bonds, the metal-chalcogen bond is much weaker due to its special sandwich structure, leading to variance in the phonon dispersions and Grüneisen parameters.[41] All of these should be considered for thermal management in the design of nanoelectronic devices based on TMD materials.

Among the semiconducting TMDs, the thermal properties of Mo- and W-based materials are widely studied. Theoretically, the transition metal atom has a minimal effect on thermal conductivity for Mo- and W-based TMDs in the 2H structure and the thermal conductivity is mainly dominated by the chalcogen species, especially for the tellurides and selenides.[42] From high-field electrical measurement and electrothermal modeling, the thermal conductivity of semi-metallic WTe_2 is 3 W m^{-1} K^{-1}[43] while density functional theory (DFT) calculation yields a higher value of around 19 W m^{-1} K^{-1} for $MoTe_2$ and WTe_2. The thermal conductivity of single-layer and bilayer $MoSe_2$ is 59 ± 18 and 42 ± 13 W m^{-1} K^{-1}, respectively, measured by optothermal Raman technique,[44] which is comparable to the value calculated by classical nonequilibrium molecular dynamics (NEMD) simulations.[45] While WS_2 shows a much larger thermal conductivity estimated from first-principles calculations than other S-based TMDs, due to a relatively large atomic weight difference induced phonon bandgap and suppressed

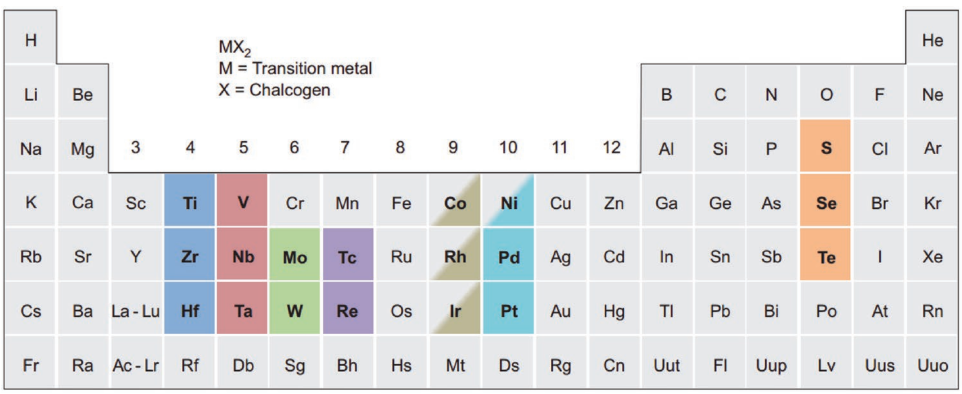

Figure 2. Transition metals and chalcogen elements forming TMDs are highlighted in the periodic table. Reproduced with permission.[31] Copyright 2013, Nature Research.

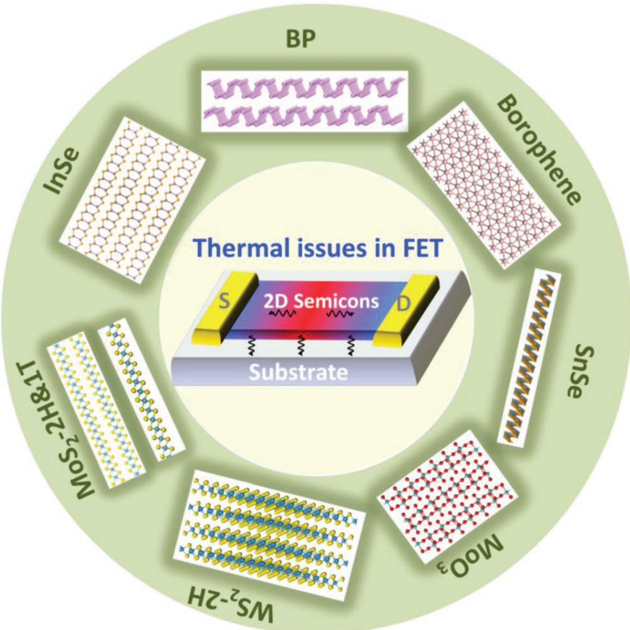

图1. 应用于场效应晶体管器件的代表性二维半导体。展示了多种二维半导体，如1T-和2H-MoS_2、BP、InSe和硒化亚锡。

金属和跨此类界面的传输，包括MoS_2/接触和MoS_2/基底，将在第3节中讨论。除了FET热管理中的这些主要考虑因素外，第4节还将讨论与二维材料其他器件应用相关的热现象，即热电效应、光电现象中的电子-声子耦合、电场以及应变对热性能和热功能器件（如热晶体管）的影响。

2. 二维半导体的热特性

2.1. 过渡金属二硫属化物

石墨烯由于缺乏电带隙 [1] 而在电子器件应用方面受到限制，这促使人们寻找具有半导体特性的类似二维材料。过渡金属二硫属化物（化学式为MX_2，其中M代表过渡金属元素，X代表硫族元素，如 **图2** 所示）是极具潜力的候选材料。当材料厚度从块体材料减薄至单层结构时，这类化合物会经历从间接带隙到直接带隙的转变，从而为电子器件和光电子器件的应用开辟了新途径。单层过渡金属二硫属化物具有约1-2电子伏特的半导体带隙，而对应的块体材料则展现出截然不同的特性——从绝缘体、半导体到半金属乃至真正金属。[31]

由于过渡金属二硫属化物（TMDs）与石墨烯在晶格结构上的差异，这类材料展现出截然不同的声子传输特性。理论计算与实验测量均表明，TMDs的热导率显著低于石墨烯。与石墨烯热导率随尺寸变化的特性不同，TMDs的热导率通常几乎不受尺寸和表面粗糙度的影响，这主要归因于其声子平均自由程（MFP）极短的固有特性。[41] 此外，由于金属-硫族元素键具有特殊的三明治结构，其键合强度远弱于强的sp^2碳-碳共价键，这导致声子色散关系和格林艾森参数存在显著差异。在基于TMD材料设计纳米电子器件时，[41] 这些特性都需要纳入热管理设计的考量。

在过渡金属硫属化合物（TMDs）中，钼基和钨基材料的热学特性研究最为广泛。理论研究表明，在二硫化物结构的钼基和钨基TMDs中，过渡金属原子对热导率的影响微乎其微，其热导率主要由硫族元素决定，尤其是碲化物和硒化物。[42] 通过高场电学测量和电热建模发现，半金属钨碲$_2$的热导率$^{-1}$ K^{-1}为3W·m^{-1} K^{-1} [43]，而密度泛函理论（DFT）计算显示钼碲$_2$和钨碲$_2$的热导率更高，约为19W·m^{-1} K^{-1}。采用光热拉曼技术测得单层和双层硒化钼$_2$的热导率分别为59±18W·m^{-1} K^{-1}和42±13W·m^{-1} K[44]，该数值与经典非平衡分子动力学（NEMD）模拟计算结果高度吻合。[45] 尽管WS_2的第一性原理计算热导率远高于其他硫族过渡金属二硫化物（TMDs），但其较大的原子量差异导致的声子带隙效应和抑制作用

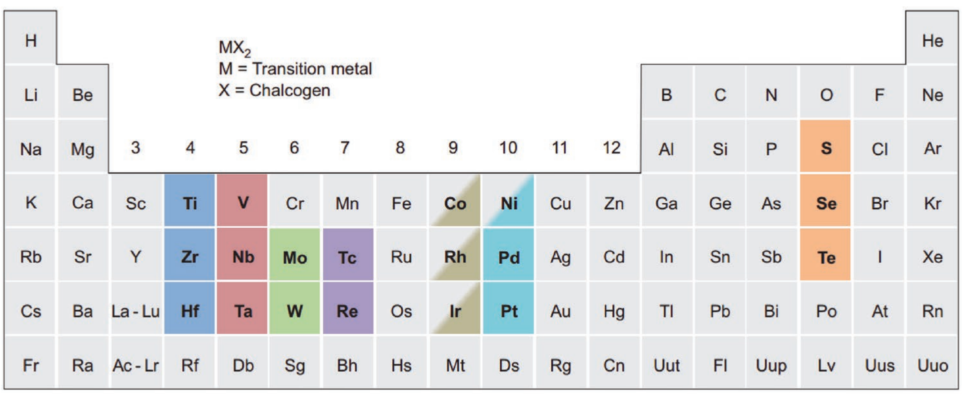

图2. 金属元素和硫族元素形成TMDs的元素在周期表中被突出显示。经许可转载。[31] 版权所有2013年，自然研究。

possibility of phonon–phonon scattering. This results in long phonon relaxation time.[46]

As one of the most stable TMDs, MoS_2 has been experimentally fabricated by various approaches, and thermal transport in this material has been widely explored. Nevertheless, the thermal conductivity values reported by different research groups diverge greatly. From ballistic NEGF calculations, the thermal conductivity of MoS_2 nanoribbons has been shown to have a strong dependence on their orientation due to the different edge effect, and reported to be 673.6 and 841.1 W m^{-1} K^{-1} for armchair and zigzag directions, respectively.[47] Later on, first-principles-based Peierls–Boltzmann transport equation (PBTE) method predicted much lower thermal conductivity[48] that is comparable to the experimental values. To further minimize the thermal conductivity, Ding et al. proposed MoS_2-$MoSe_2$ lateral superlattice structure,[49] which showed close to 80% reduction in thermal conductivity compared with that of individual single-layer due to the enhanced anharmonic phonon scattering. On the other hand, even though it was theoretically predicted that the number of layers leads to different thermal conductivity as a result of the change in phonon dispersion and phonon scattering anharmonicity, the experimental data show much weaker layer dependence (**Figure 3**). The thermal conductivity is 34.5 ± 4,[50] 77 ± 25,[44] and 52 W m^{-1} K^{-1},[51] for 1-layer (1-L), 2L, and 11L MoS_2, measured by noncontact Raman spectrometry, where temperature-sensitive Raman peaks are commonly employed to deduce the thermal conductivity of 2D layered materials. Both thermal bridge method[52] and e-beam technique[53] give similar thermal conductivity values for 4L MoS_2, which are much smaller than that of bulk MoS_2 obtained by time-domain thermoreflectance (TDTR) technique.[54] These experimental discrepancies are primarily attributed to the sample quality, such as extra PMMA induced surface disorders, as well as experimental measurement inaccuracies, which inevitably obfuscate the intrinsic thermal conductivity of MoS_2. For detailed descriptions of the measurement setups, the reader may refer to other review papers.[37,38]

Isotopic defects could affect the phonon transport by increasing phonon scattering.[55,56] More recently, the isotope effect on thermal transport in monolayer MoS_2 has been investigated,[57] where the thermal conductivity of isotopically pure Mo $^{100}MoS_2$ and 50% $^{100}MoS_2$ was found to be higher by 50% and 30%, respectively, compared to that of natural MoS_2 comprising naturally abundant isotopes of Mo of various atomic masses of 92, 94, 95, 96, 97, 98, and 100.[57,58] Due to the reduction of the isotopic disorder, the isotopically pure MoS_2 shows less isotope-phonon scattering, which enhances the phonon MFPs and thus the final thermal conductivity. These results are shown in **Figure 4**.

For other members having semiconducting property, like group IVB and VIII transition metal based TMDs, a much lower thermal conductivity has been theoretically shown, which is beneficial for TE applications.[60] Typically, the thermal conductivity of the TMDs from these groups has a similar dependence on the chalcogen atom species and increases from selenides to sulfides. The thermal conductivity of monolayer $ZrSe_2$ and $HfSe_2$ at room temperature is 1.2 and 1.8 W m^{-1} K^{-1},[61] respectively, which is lower than that of monolayer ZrS_2, of around

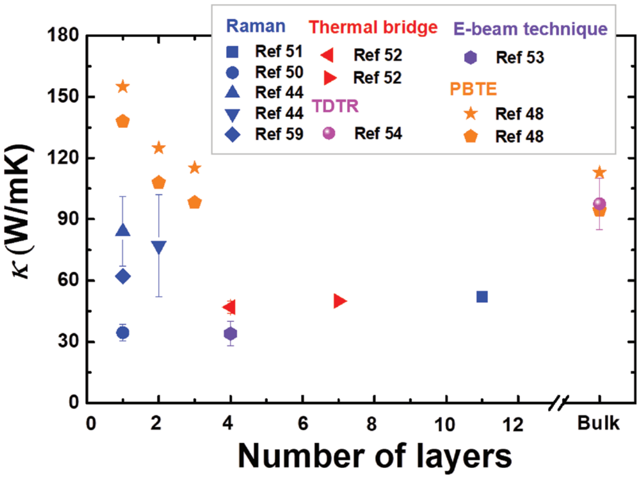

Figure 3. Thermal conductivity of MoS_2 as a function of layers numbers at room temperature. The κ is measured by various experimental techniques, including Raman spectrometer,[44,50,51,59] thermal bridge method,[52] e-beam heating technique,[53] time-domain thermoreflectance (TDTR),[54] and theoretical calculation by PBTE.[48]

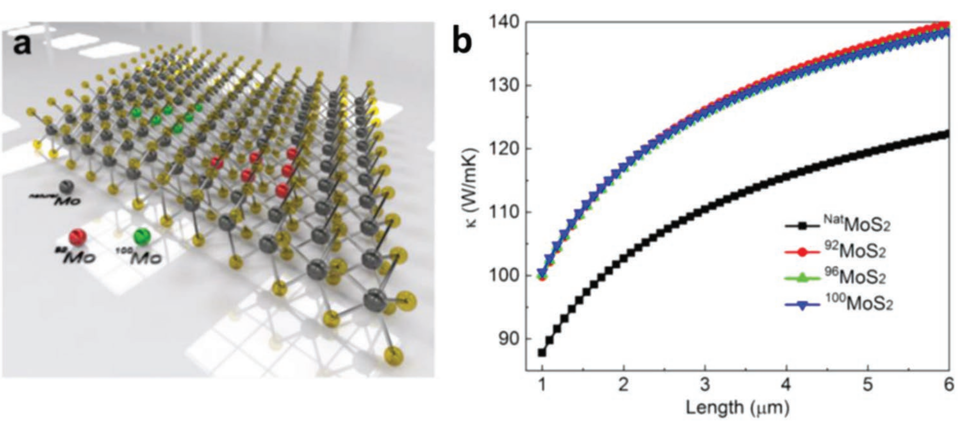

Figure 4. a) A sketch of different Mo isotoped-MoS_2. b) The sample length-dependent thermal conductivity of natural-MoS_2, $^{92}MoS_2$, $^{96}MoS_2$, and $^{100}MoS_2$. Reproduced with permission.[57] Copyright 2019, American Chemical Society.

声子-声子散射的可能性。这导致了声子弛豫时间的延长。[46]

作为最稳定的过渡金属硫属化合物之一，MoS_2 已通过多种方法实现实验制备，其热传导特性也得到广泛研究。然而，不同研究团队报告的热导率值存在显著差异。根据弹道NEGF 计算，MoS_2 纳米带的热导率因其边缘效应不同而对取向高度敏感，分别在扶手椅型和锯齿型取向下测得673.6和841.1 $W \cdot m^{-1} \cdot K^{-1}$。[47] 后来，基于第一性原理的皮尔尔斯-玻尔兹曼输运方程（PBTE）方法预测的热导率 [48] 与实验值相近。为进一步降低热导率，丁等人提出了MoS_2-$MoSe_2$ 横向超晶格结构，[49] 该结构通过增强非谐声子散射使热导率较单层材料降低近80%。另一方面，尽管理论预测层数变化会导致声子色散和声子散射非谐性改变，从而产生不同的热导率，但实验数据表明其层依赖性要弱得多（**图3**）。通过非接触式拉曼光谱测量的热导率分别为34.5±4、[50] 77±25 [44] 和52 $W \cdot m^{-1} \cdot K^{-1}$ [51]（对应单层、双层和11层MoS_2），而温度敏感的拉曼峰通常被用于推断二维层状材料的热导率。无论是热桥法 [52] 还是电子束技术 [53]，测得的4层MoS_2 热导率值均远低于通过时域热反射（TDTR）技术获得的块体MoS_2 热导率。[54] 这些实验差异主要归因于样品质量，例如额外的PMMA 诱导表面紊乱，以及实验测量的不准确性，这些不可避免地模糊了MoS的本征热导率 $_2$。关于测量设置的详细描述，读者可参考其他综述论文。[37 ， 38]

同位素缺陷可能通过增加声子散射来影响声子输运。[55,56] 最近，单层MoS中同位素效应对热输运的影响 $_2$ 已被研究。

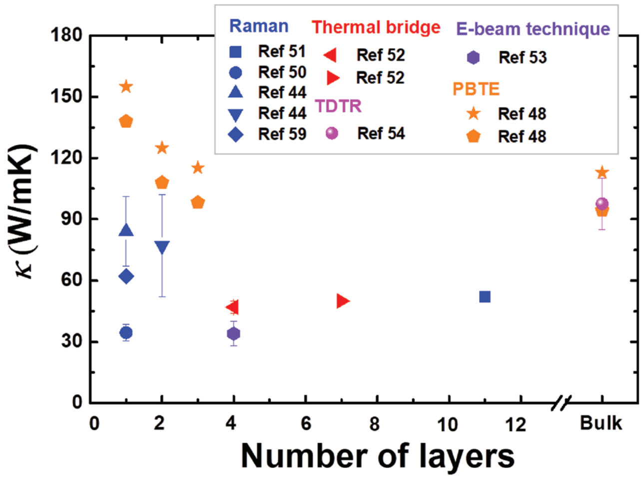

图3. MoS_2 的热导率随层数变化的室温曲线。κ 通过多种实验技术测量，包括拉曼光谱仪、[44,50,51,59] 热桥法、[52] 电子束加热技术、[53] 时域热反射（TDTR）、[54] 以及通过 PBTE 进行的理论计算。[48]

研究发现，[57] 同位素纯Mo $^{100}{}_2$ MoS $_2$ 和50% ^{100} MoS的热导率分别比天然MoS高50%和30%，后者 $_2$ 含有天然丰富的92、94、95、96、97、98和100号原子质量的Mo同位素。[57,58] 由于同位素无序度降低，同位素纯MoS_2 显示出更少的同位素-声子散射，这增强了声子平均自由程，从而提高了最终热导率。这些结果如 **图4** 所示。

对于其他具有半导体特性的成员，如基于IVB族和VIII族过渡金属的过渡金属硫属化合物（TMDs），理论研究表明其热导率要低得多，这对热电应用是有利的。[60] 通常，这些组的TMDs热导率对硫族原子种类的依赖性相似，并且从硒化物到硫化物逐渐增加。单层$ZrSe_2$ 和$HfSe_2$ 在室温下的热导率分别为1.2和1.8 $W \cdot m^{-1} \cdot K^{-1}$，[61] 分别低于单层$ZrS_2$ 的约

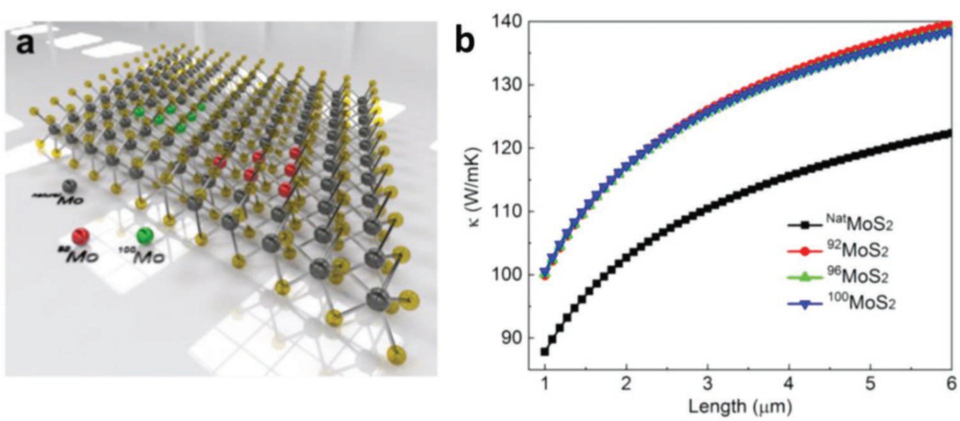

图4. a) 不同Mo同位素-MoS_2 的示意图。b)天然-MoS_2、$^{92}MoS_2$ $^{96}{}_2$ 和 $^{100}MoS_2$ 的样品长度依赖热导率。经许可转载。[57] 版权所有2019年美国化学会。

3.2 W m^{-1} K^{-1}.[62] Similarly, the thermal conductivity of monolayer PdS_2[63] is much larger than that of monolayer $PdSe_2$.[64] It may be noticed that all these values are from theoretical calculations and more experimental work needs to be carried out to further explore the thermal transport behaviors of these TMDs.

More interestingly, unlikely the isotropic thermal transport in the in-plane direction of most TMDs, ReS_2 in a distorted 1T phase shows strong anisotropic thermal conductivity along and transverse Re-chains,[65] which is similar to the thermal transport in puckered structures discussed below. Monolayer $PdSe_2$, due to its puckered morphology, shows direction-dependent thermal transport along its in-plane direction as well and the thermal conductivity of $PdSe_2$ along the x- and y-directions is 3.7 (1.4) and 7.2 (2.7) W m^{-1} K^{-1} at 300 K (800 K), respectively. This particular property could provide an additional degree of freedom in the design of nanoscale thermal devices based on these materials.

2.2. Phosphorene

Phosphorene, a single layer of phosphorus atoms arranged in a honeycomb-like structure, has drawn a great deal of attention as a promising 2D semiconductor due to its layer-tunable bandgap and high carrier (electron, hole) mobility, which is an important consideration for potential candidates for modern electronic applications, i.e., TE devices,[66,67] photodetectors,[68] and field-effect transistors.[69,70] Phosphorene has two natural allotropes–blue phosphorene and black phosphorene,[71,72] which show different lattice structures and thus have different phonon transport properties. Because of its special puckered honeycomb structure, black phosphorene exhibits interesting orientation-dependent phonon transport properties.[73] As blue phosphorene is isotropic with a zigzag structure, phonon transport in this material does not depend on the orientation in the in-plane direction.[71]

Typically, few-layer phosphorene is exfoliated from crystalline phosphorus by means of mechanical cleavage, and other techniques such as plasma-assisted process[74] and liquid-phase fabrication technique[75] have been proposed as well. **Figure 5**a shows the atomic structure of black phosphorene, stacked together into black phosphorene by van der Waals force, and each atom is bonded to three neighboring atoms in sp[3] hybridization configuration.[76] Theoretically, it was predicted that phonon transport along zigzag (ZZ) direction is preferred compared to armchair (AC) direction and many first principles calculations have investigated this interesting anisotropic thermal transport,[71,77–79] which is attributed to structural-asymmetry-induced phonon group velocity and relaxation time. It was demonstrated as well that the strain would have an inconsistent effect on the thermal transport along these two

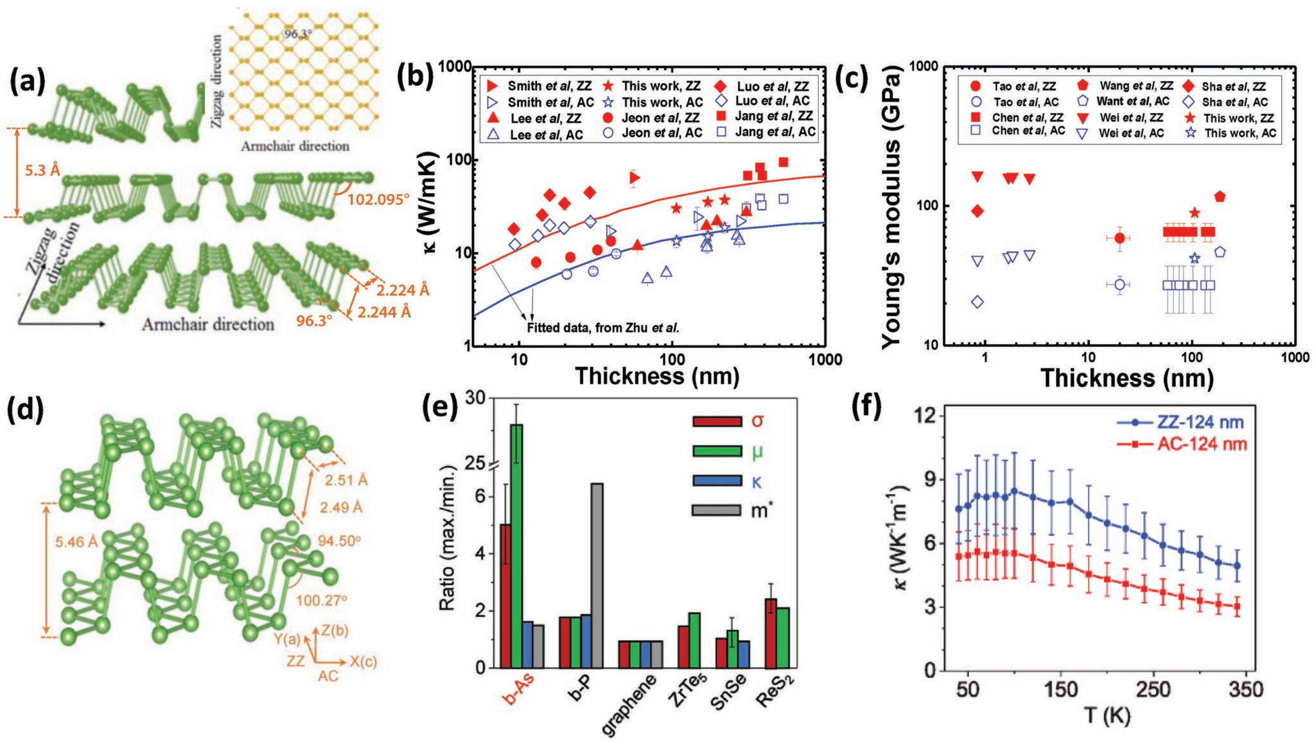

Figure 5. a) Atomic structure of multilayer black phosphorus with top view of monolayer phosphorene shown in the inset. Reproduced with permission.[76] Copyright 2016, Wiley-VCH. b) A summary of thickness dependent thermal conductivity of black phosphorene with various dimensions, like black phosphorene thin films, nanosheets and NRs. c) Thickness dependent Young's modulus, including both theoretical calculations and experimental measurement. Reproduced with permission.[91] Copyright 2018, Wiley-VCH. d) 3D sketch of b-As structures. e) Comparison of in-plane anisotropy ratio of various properties along the AC and ZZ directions of b-As and other 2D materials. b-As has shown the maximum anisotropy ratio in both electrical conductivity and mobility while comparable thermal conductivity with other materials. f) Temperature-dependent thermal conductivity of b-As nanoribbons with dimension ≈124 nm along the ZZ and AC directions. Reproduced with permission.[93] Copyright 2018, Wiley-VCH.

3.2 在 $^{-1}$ K $^{-1}$[62] 中，单层PdS_2[63]的热导率远大于单层$PdSe_2$[64]。值得注意的是，所有这些数值均来自理论计算，需要开展更多实验工作来进一步探索这些TMDs的热输运行为。

更有趣的是，大多数过渡金属二硫属化物（TMDs）在面内方向表现出各向同性的热传导特性，而处于扭曲一维（1T）相的铼化硒（ReS_2）却展现出沿Re链方向和横向的显著各向异性热导特性，[65] 这与下文讨论的褶皱结构热传导现象具有相似性。单层硒化钯（PdSe）$_2$ 由于其褶皱结构，同样呈现面内方向的各向异性热传导特性。在300K（800K）温度下，$PdSe_2$ 在 x 方向和 y 方向的热导率分别为3.7（1.4）和7.2（2.7）W· m^{-1} · K^{-1} 。这一独特性质为基于此类材料设计纳米级热器件提供了额外的设计自由度。

2.2. 磷烯

磷烯是一种由磷原子单层构成的蜂窝状结构材料，因其能层可调的带隙和高载流子（电子、空穴）迁移率，作为有前景的二维半导体材料而备受关注。磷烯是现代电子器件（如热电元件、[66,67] 光电探测器 [68] 和场效应晶体管）潜在候选材料的重要考量对象。[69,70] 该材料存在两种天然同素异形体——蓝色磷烯与黑色磷烯，[71,72] 其晶格结构差异导致声子传输特性迥异。黑色磷烯凭借独特的褶皱蜂窝结构，展现出显著的方向依赖性声子传输特性。[73] 而蓝色磷烯因具有锯齿状结构呈现各向同性，其面内方向的声子传输特性不受取向影响。[71]

通常，少层磷烯是通过机械剥离法从结晶磷中制备的，此外还提出了等离子体辅助工艺 [74] 和液相制备技术 [75] 等其他方法。**图5a** 展示了黑磷烯的原子结构：原子间通过范德华力堆叠形成黑磷烯结构，每个原子以sp [3] 杂化构型与三个相邻原子键合。[76] 理论预测表明，相较于扶手椅型排列，沿锯齿形（ZZ）方向的声子传输更为有利，大量基于第一性原理的计算研究揭示了这种独特的各向异性热传导特性，[71,77–79] 其根源可追溯至结构不对称引发的声子群速度与弛豫时间差异。研究还证实，应变对这两种热传导方向的影响存在显著差异。

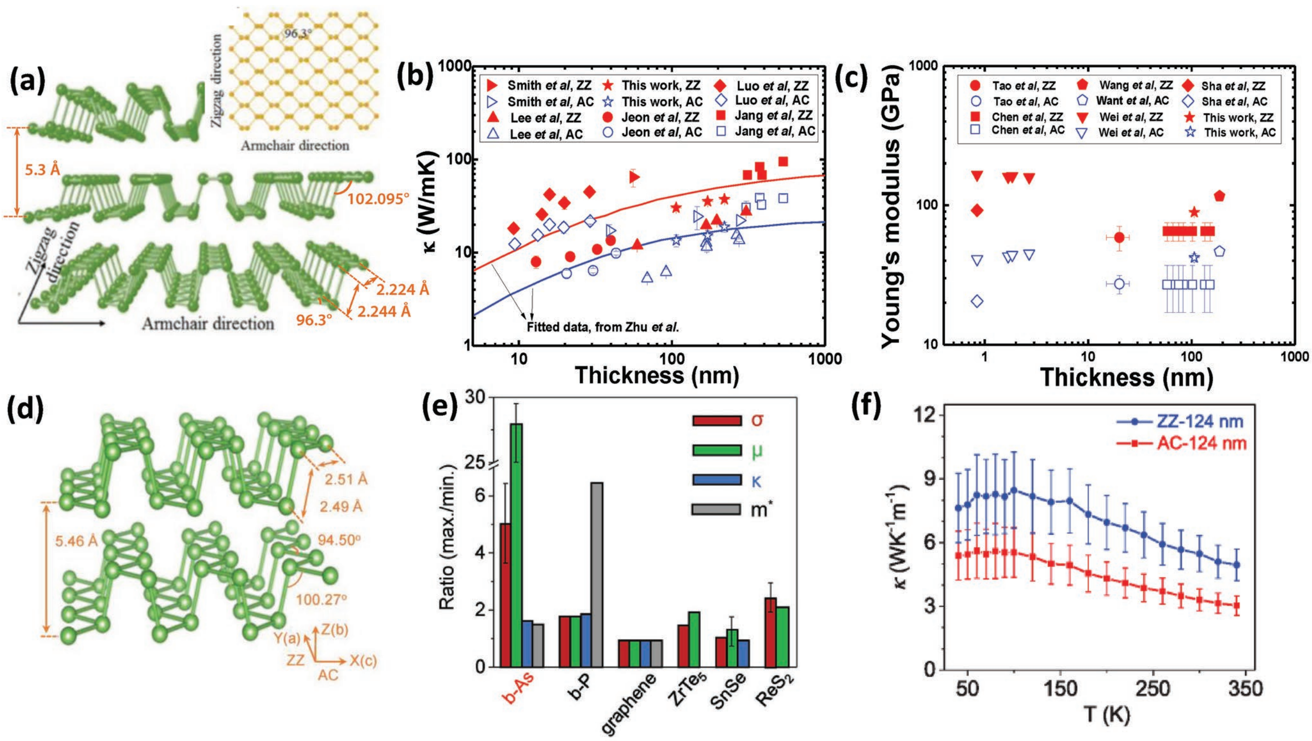

图5. a)多层黑磷的原子结构示意图，插图为单层磷烯俯视图。经授权转载。[76] 版权所有2016年Wiley- VCH 。b)不同尺寸黑磷烯（如薄膜、纳米片和纳米带）厚度依赖热导率的总结。c)杨氏模量随厚度变化的对比，包含理论计算与实验测量数据。经授权转载。[91] 版权所有2018年Wiley- VCH 。d)β-As结构的三维示意图。e)β-As与其他二维材料在AC和ZZ方向上各物理量面内各向异性比的对比。β-As在电导率和迁移率方面均呈现最大各向异性比，同时热导率与其他材料相当。f)尺寸约124纳米的β-As纳米带在ZZ和AC方向上的温度依赖热导率。经授权转载。[93] 版权所有2018年Wiley- VCH 。

directions. By using nonequilibrium Green's functions method combined with first principles calculation, Ong et al.[80] showed that the ZZ-oriented thermal conductance is increased when a strain is applied along this direction and decreases when an AC-oriented strain is applied; while the thermal conductance in AC orientation is always reduced when a strain is applied to either ZZ- or AC-directions. This inconsistent phonon response to strain possibly explains the recent experimental observation that Raman sensitive peak A_g^1, B_{2g}, and A_g^2 modes could have opposite responses for tensile strain ZZ- and AC-directions.[81] By using a polymeric low-k dielectric to stretch the substrate, a high tensile strain of more than 7% was achieved in ref. [81], which nonetheless is still short of the theoretical predication of the high strain limit for ZZ and AC directions of 27% and 30%,[82] respectively. Interestingly, the in-plane modes, B_{2g} and A_g^2, show a red/blue shift when a strain is introduced along the ZZ and AC directions, respectively, due to the special in-plane puckered structure of BP. Even though the out-of-plane mode, A_g^1, remains unchanged for ZZ direction strain, it shows red shifting for AC direction strain. This study of strained BP provides insight into strain engineering of BP for electronic and optical devices.[83,84]

Experimentally, different measurement techniques have been employed to explore the thermal transport anisotropy behaviors in black phosphorene, such as the use of suspended-pad thermal bridge,[85,86] TDTR[87,88] and micro-Raman spectroscopy,[89] and the measurement results and sample dimensions are summarized in **Table 1**, where the anisotropy ratio are compared for different techniques as well. According to Zhu et al.,[90] the strong anisotropy of thermal transport in black phosphorene is attributed to both group velocity variations and relaxation time variations along different crystalline orientations. The role of phonon relaxation time for thermal transport anisotropy is doubtful since dominant direction-dependent phonon dispersion[85] and similar intrinsic phonon scattering rate along ZZ and AC directions[89] were demonstrated. More recently, Zhao et al.[91] decoupled the dominant role of phonon group velocity from phonon relaxation time by linking the anisotropic phonon group velocity to anisotropic sound velocity, as verified by measuring the Young's modulus of black phosphorene along both ZZ and AC directions using method of three-point bending. It turns out that the anisotropy ratio between κ_{ZZ} and κ_{AC} is comparable to that of the Young's modulus value along these two directions. Considering the fact that the speed of sound, v_s^2 is proportional to the Young's modulus E and that the phonon group velocity could be treated as the speed of sound in the low frequency regime, the anisotropic phonon dispersion relation therefore accounts for the anisotropy observed in thermal transport between ZZ and AC directions. Further change of anisotropy ratio of thermal conductivity between ZZ and AC directions can be realized by the introduction of "nonsquare" pores in a phononic crystal (PnC),[92] where phonon group velocity along the direction of more porous areas would be degraded more severely. This adjustable anisotropic ratio of thermal transport is a unique property of 2D phosphorene that can be exploited in thermal management applications.

As a cousin of black phosphorene, black arsenic (b-As) has just been fabricated and shows interesting anisotropic transport behaviors in electrical and thermal properties.[93,94] Its puckered atomic structure, shown in Figure 5d, is similar to that of black phosphorene. Among the materials with puckered lattice structures, such as black phosphorene and other puckered materials, b-As has the greatest anisotropy in both electrical conductance and carrier mobility as shown in Figure 5e. For b-As, the electrical transport is more favorable in the AC direction than that in ZZ direction while thermal conductance just behaves in the opposite trend, similar to that of black phosphorene, which thus enables the implementation of novel transverse TE devices.[93] Nevertheless, unlike BP which easily oxidizes, b-As demonstrates good resistance to ambient degradation.[94]

Table 1. Thermal conductivity of BP at room temperature.

Dimension	Method	κ [W m^{-1} K^{-1}] [300 K]		Anisotropy ratio
		AC	ZZ	ZZ/AC
Monolayer[95]	First-principles calculations	4.59	15.33	3.34
Monolayer[71]	First-principles calculations	36	110	3.06
Monolayer[78]	First-principles calculations	27.8	48.9	1.76
Monolayer[96]	MD simulation	63.6	110.7	1.74
Monolayer[73]	MD simulation	33.0	152.7	4.63
Monolayer[77]	First-principles calculations	24.3	83.5	3.44
Film (13–48 nm)[86]	Suspended-pad microdevices	5.8–9.6	7.8–13.2	1.34–1.36
NRs (60–300 nm)[85]	Suspended-pad microdevices	5.4–15.5	11.7–27	1.74–2.17
Film (9–30 nm)[89]	Micro-Raman spectroscopy	12.4–22	18.2–45.1	1.5–2
Film (39–277 nm)[97]	Four-probe measurement	17.3–24.5	64.9	≈3
NRs (106–220 nm)[91]	E-beam technique	13.5–20.6	30.5–38.6	1.99–2.33
Film (138–552 nm)[87]	TDTR	26.4–33.9	63–86.4	2.39–2.55
Bulk[88]	TDTR	28 ± 5	83 ± 10	≈2.96
Bulk[90]	TR-MOKE	26–36	84–101	≈3

翁等人[80]通过结合非平衡格林函数方法与第一性原理计算发现：当沿ZZ方向施加应变时，热导率会增强；而沿AC方向施加应变时则会降低。值得注意的是，无论应变方向是ZZ还是AC，AC方向的热导率始终呈现下降趋势。这种声子响应的差异性，或许能解释近期实验观测到的拉曼敏感峰A_g^1、B_{2g}和A_g^2模式在ZZ与AC方向拉伸应变下会产生相反响应的现象。[81]在参考文献[81]中，研究者采用聚合物低 *k* 介电材料对基底进行拉伸，成功实现了超过7%的高应变值，但仍[82]未达到理论预测的ZZ和AC方向高应变极限值—— $_{27\%}$ 和 $_{30\%}$。有趣的是，当沿ZZ和AC方向施加应变时，BP材料特有的面内褶皱结构使其面内模式B_{2g}和A_g^2会分别出现红移和蓝移现象。尽管面外模式A_g^1在ZZ方向应变下保持不变，但在AC方向应变时却呈现红移特征。这项应变BP研究为电子和光学器件中的应变工程提供了重要参考。[83,84]

实验研究中，科研人员采用多种测量技术探究黑磷烯的热输运各向异性特性，包括悬垫式热桥法、[85,86] TDTR [87,88]和微拉曼光谱法。[89]相关测量结果及样品尺寸详见**表1**，其中不同技术手段测得的各向异性比值也进行了对比分析。朱等人指出，[90]黑磷烯热输运的显著各向异性源于晶向不同导致的群速度差异与弛豫时间变化。不过关于声子弛豫时间对热输运各向异性的具体作用仍存争议，因为研究已证实沿ZZ和AC方向[89]存在主导的各向异性声子色散现象，且这两种方向的本征声子散射率基本一致。

最近，赵等人[91]通过将各向异性声子群速度与各向异性声速联系起来，成功将声子群速度主导作用与声子弛豫时间分离。这一结论通过三点弯曲法测量黑磷烯在κ_{ZZ}和κ_{AC}方向的杨氏模量得到验证。研究发现，$_{\kappa}$ZZ与κ AC之间的各向异性比值与这两个方向的杨氏模量值相当。考虑到声速v_s^2与杨氏模量 *E* 成正比，且在低频区域声子群速度可视为声速，因此各向异性声子色散关系能够解释ZZ与AC方向间热输运观测到的各向异性现象。通过在声子晶体（PnC）中引入“非方形”孔隙结构，可进一步调控热传导率在ZZ和AC方向上的各向异性比。[92]这种结构会使多孔区域的声子群速度沿特定方向的衰减更为显著。二维磷烯材料特有的可调各向异性热传导特性，为热管理应用提供了独特解决方案。

作为黑磷烯的近亲，黑砷（b-As）最近被成功制备，展现出独特的电学与热学各向异性传输特性。[93,94]其如图5d所示的褶皱原子结构与黑磷烯高度相似。在黑磷烯等褶皱晶格材料中，b-As展现出最大的电导率和载流子迁移率各向异性，如图5e所示。相较于ZZ方向，b-As在AC方向的电传输更为有利，而热导率则呈现相反趋势，这与黑磷烯的特性如出一辙，为开发新型横向热电器件提供了可能。[93]不过与易氧化的黑磷烯不同，b-As表现出优异的环境稳定性，不易发生退化。[94]

表1. BP在室温下的热导率。

维度	方法	*κ [W m⁻¹ K⁻¹] [300 K]*		各向异性比
		AC	Z字形	ZZ/AC
单层[95]	第一原理计算	4.59	15.33	3.34
单层[71]	第一原理计算	36	110	3.06
单层[78]	第一原理计算	27.8	48.9	1.76
单层[96]	分子动力学模拟	63.6	110.7	1.74
单层[73]	分子动力学模拟	33.0	152.7	4.63
单层[77]	第一原理计算	24.3	83.5	3.44
薄膜（13-48 nm）[86]	悬浮垫微器件	5.8–9.6	7.8–13.2	1.34–1.36
NRs（60–300 nm）[85]	悬浮垫微器件	5.4–15.5	11.7–27	1.74–2.17
薄膜（9–30 nm）[89]	微拉曼光谱	12.4–22	18.2–45.1	1.5–2
薄膜（39–277 nm）[97]	四探头测量	17.3–24.5	64.9	≈3
NRs（106–220 nm）[91]	电子束技术	13.5–20.6	30.5–38.6	1.99–2.33
薄膜（138-552 nm）[87]	TDTR	26.4–33.9	63–86.4	2.39–2.55
批量[88]	TDTR	28 ± 5	83 ± 10	≈2.96
批量[90]	TR-MOKE	26–36	84–101	≈3

2.3. Other 2D Semiconductors

Other than the TMDs and BP discussed above, the elements in Groups III to VI have drawn a lot of interest as well for interesting 2D layered semiconducting materials. Among them, the compounds of indium selenide, such as InSe and $InSe_2$ and compounds of bismuth telluride, such as Bi_2Te_3, Bi_2Se_3, and Sb_2Te_3 are the representative ones in various applications in optoelectronic devices,[98] high-mobility filed-effect transistors,[99] phase-change memory,[100] single-mode phonon laser,[101] and TE.[102] The thermal transport in these materials has been studied quite recently. From EMD calculation, Bi_2Te_3 exhibits bulk-like thermal conductivity value when its thickness is around 5 nm, and the thermal conductivity has a nonmonotonic dependence on its thickness due to the interplay between Umklapp scattering and boundary scattering.[103] Later experimental work by suspended pads measurement shows that the thermal conductivity increases with the thickness,[102] as shown in **Figure 6**a, where the phonon-boundary scattering is dominant in the phonon transport.

Different from Bi_2Te_3 that has shorter phonon MFPs, InSe has relatively larger MFPs. The multiple crystalline phases of few-layered In_2Se_3 sheets are stable under experimental conditions,[104] which provides the possibility of conducting further experimental study on other physical properties. In 2016, Zhou et al. measured the thermal conductivity of single-crystal In_2Se_3 sheets.[105] In_2Se_3 powders were used in the CVD growth of 2D In_2Se_3 sheets, and H_2 was used as the carrier gas. Through the temperature dependence of the phonon modes, thermal conductivity was measured using micro-Raman spectroscopy method. In general, in-plane thermal conductivity of In_2Se_3 sheets suspended across holes increases with increasing layer thickness in the range from 5 to 35 nm. Specifically, in this range of thickness, in-plane thermal conductivity increases from ≈4 to ≈60 W m^{-1} K^{-1}. This thickness dependence is similar to that observed in supported graphene, but opposite that in suspended graphene sheets[106] as shown in Figure 6a. This suggests the dominant role of phonon-boundary scattering in the thermal transport process in In_2Se_3, instead of phonon–phonon anharmonic scattering.

Other layered semiconducting materials belonging to these families have been proposed for TE applications considering their ultralow thermal conductivity, such as $GeAs_2$[107] and SnSe.[108] Since the dimensionless figure of merit ≈2.62 was experimentally reported,[109] layered SnSe has been considered as an excellent candidate for commercial development. Natural SnSe has a layered orthorhombic crystal structure and its monolayer has a highly puckered structure, very similar to that of puckered BP.[109–111] Zhang et al. proposed a phase-controlled method to synthesize SnSe nanosheets[112] and a single-crystalline SnSe nanosheet with thickness of ≈1.0 nm and lateral size of ≈300 nm was synthesized experimentally.[113] Using density functional theory combined with Boltzmann transport theory, Wang et al. studied thermal and thermoelectric properties of single-layered SnSe.[108] It was demonstrated that anharmonic phonon–phonon interaction is dominant in the phonon scattering process, because of the remarkable overlapping between low frequency optical branches with the acoustic branches, as a consequence of the heavy atomic masses, weak interatomic bonds and the low atomic coordination. At room temperature, in addition to the contributions of acoustic branches to thermal conductivity, the low frequency optical modes also have nontrivial contributions. Thus, due to an anomalously high Grüneisen value and high anharmonicity of chemical bonds, there is strong anharmonicity in phonon–phonon scattering process, which leads to a low thermal conductivity (3.0 W m^{-1} K^{-1} for ZZ direction and 2.6 W m^{-1} K^{-1} for AC direction) in monolayer SnSe according to first-principles calculations.[114] This strong phonon anharmonicity is theoretically attributed to the long-range interaction driven by the resonant bonding.[95,115–117] The thermal conductivity of bulk SnSe is smaller than that of monolayer SnSe possibly due to the smaller interlayer interactions induced smaller phonon group velocity, as shown in Figure 6b, which thus contributes to better TE property considering its ultralow thermal conductivity (experimentally measured ≈0.25–0.28 W m^{-1} K^{-1} at 773 K).[109] The strong phonon anharmonicity was later demonstrated by inelastic neutron scattering (INS) measurement,[118] where a softer transverse acoustic (TA) and transverse optic (TO) mode along Γ-X direction was observed than that of Γ-Y direction and this accounts for its anisotropic thermal transport as well.[110] Unlike the weak direction dependence in thermal conductivity of SnSe, anisotropic and low thermal conductivity was presented in $GeAs_2$, a semiconductor formed from group IV and V elements. At room

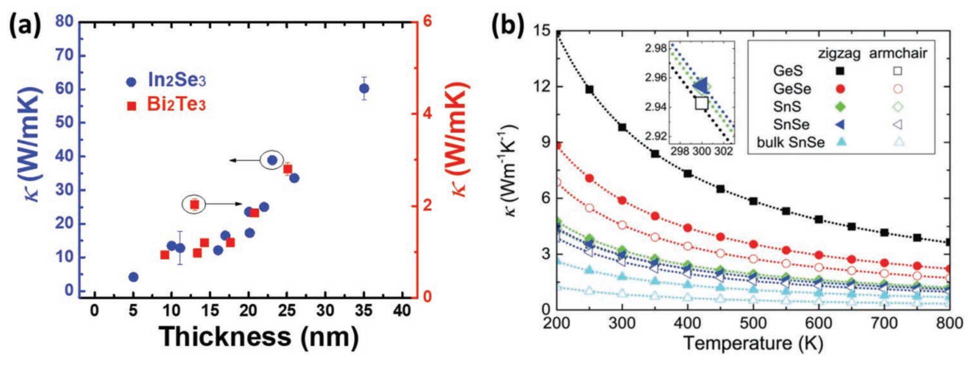

Figure 6. a) Thickness dependent thermal conductivities of In_2Se_3[105] and Bi_2Te_3[102] by experimental measurement. A weak dimension dependent for Bi_2Te_3 is shown. b) Temperature-dependent thermal conductivity of monolayer GeS, GeSe, SnS and SnSe, and bulk SnSe. Reproduced with permission.[114] Copyright 2016, Royal Society of Chemistry.

2.3. 其他2D半导体

除了前文讨论的过渡金属二硫属化物（TMDs）和硼族化合物（BP）外，III至VI族元素因其独特的二维层状半导体特性也备受关注。其中，硒化铟（InSe）$_2$ 与碲化铋（Bi $_2$ Te $_3$）、 $_2$ 硒化铋（Bi $_{3\,2}$ Se）$_3$ 、 [100] 硫化锑（Sb $_2$ Te）等化合物在光电子器件、 [98] 高迁移率场效应晶体管、 [99] 相变存储器、 [101] 单模声子激光器 [102] 及热电材料（TE）等领域具有重要应用价值。 $_3$ 近期研究发现，铋 $_2$ 硫化物（Bi $_2$ Te）在厚度约为5纳米时展现出块体材料级别的热导率，其热导率随厚度变化呈现非单调趋势，这源于乌姆克拉普散射与边界散射之间的相互作用。[103] 后续实验工作通过悬置电极测量发现，热导率会随着厚度增加而增大， [102] 如 **图6a** 所示，此时声子传输中以声子-边界散射为主。

与Bi $_2$ Te $_3$ 具有较短的声子平均自由程不同，InSe的声子平均自由程相对较大。实验条件下，少层In $_2$ Se $_3$ 垂片的多种晶相保持稳定， [104] 为深入研究其他物理特性提供了可能。2016年，周等人测量了单晶In $_2$ Se $_3$ 垂片的热导率。研究中采用 [105] In $_2$ Se $_3$ 粉末作为载气，通过微拉曼光谱法结合声子模式的温度依赖性进行测量。总体而言，悬空在孔洞上方的In $_2$ Se $_3$ 垂片在5-35纳米厚度范围内，其面内热导率随层厚增加而提升。具体而言，在该厚度区间内，面内热导率从约4W/（m· $^{-1}$ ·K）$^{-1}$ 增加至约60W/（m· $^{-1}$ ·K）。这种厚度依赖性与支撑石墨烯中观察到的相似，但与悬浮石墨烯片相反 [106]，如图6a所示。这表明声子-边界散射在In $_2$ Se $_3$ 的热传输过程中起主导作用，而不是声子-声子非谐散射。

考虑到这些材料家族的其他层状半导体材料已被提出用于热电应用

它们的超低热导率，例如GeAs $_2$ [107]和硒化亚锡。 [108] 由于无量纲优值≈2.62的实验报道， [109] 层状硒化亚锡被视为商业开发的优秀候选材料。天然硒化亚锡具有层状正交晶系结构，其单层结构高度褶皱，与褶皱的BP结构非常相似。 [109–111] 张等人提出了一种相控制法来合成硒化亚锡纳米片 [112]，并实验合成了厚度≈1.0纳米、横向尺寸≈300纳米的单晶硒化亚锡纳米片。 [113] 采用密度泛函理论结合玻尔兹曼输运理论，王等人研究了单层硒化亚锡的热学和热电性能。[108] 研究表明，由于原子质量较大、原子间键较弱且原子配位数较低，低频光学分支与声学分支之间存在显著重叠，非谐声子-声子相互作用在声子散射过程中占据主导地位。在室温条件下，除了声学分支对热导率的贡献外，低频光学模式也具有不可忽视的贡献。因此，根据第一性原理计算，由于化学键的格里森值异常高且非谐性较强，单层硒化亚锡在声子-声子散射过程中表现出强烈的非谐性，导致其热导率较低（ZZ方向为3.0 W· $^{m^{-1}}$ ·K </sp_[95,115–117] 块体硒化亚锡的热导率低于单层硒化亚锡，这可能是由于层间相互作用较小导致声子群速度降低，如图6b所示，因此考虑到其超低热导率（实验测得约 0.25–0.28 W m $^{-1}$ K $^{-1}$ 在773 K时），其热电性能更优。 [109] 后来通过非弹性中子散射（INS）测量证实了强烈的声子非谐性， [118] 其中沿 Γ - X 方向观察到的横向声学（TA）和横向光学（TO）模式比 Γ - Y 方向更软，这也解释了其各向异性热输运现象。 [110] 与硒化亚锡热导率方向依赖性较弱不同，由IV族和V族元素形成的半导体GeAs $_2$ 展现出各向异性且低热导率的特性。在室温下

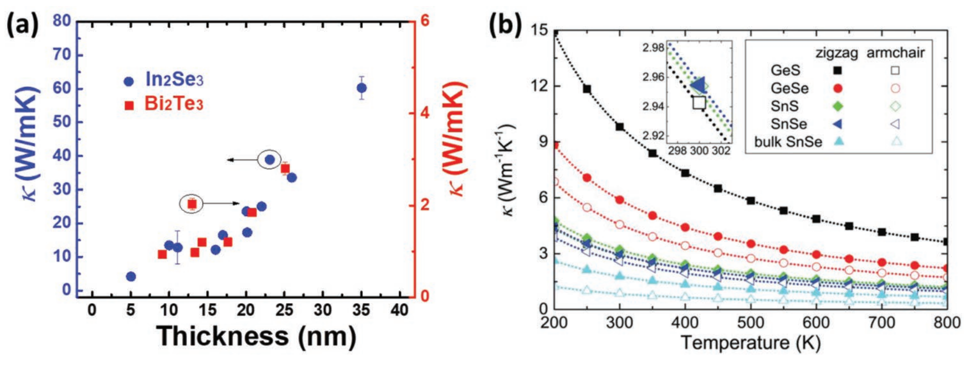

图6. a) 实验测量的In $_2$ Se $_3$ [105]和Bi $_2$ Te $_3$ [102]的厚度依赖热导率。展示了Bi $_2$ Te $_3$ 的微弱尺寸依赖性。b) 单层GeS、GeSe、SnS和硒化亚锡以及块体硒化亚锡的温度依赖热导率。经许可转载。 [114] 版权所有2016年英国皇家化学学会。

temperature, thermal conductivity of 6.03 W m^{-1} K^{-1} is reported along *a*-axis, while a much lower value of 0.68 W m^{-1} K^{-1} along the *b*-axis.[107]

Similar to the role of transition metal in thermal conductivity of TMDs, the thermal conductivity of InSe was calculated to be lower than that of GaSe due to the enhancement in acoustic phonon velocity and in the reduction of scattering between acoustic and optical phonons.[119–121] Moreover GaS has an overall larger thermal conductivity considering its even higher acoustic phonon velocity and weaker coupling of optical phonon modes with acoustic phonons.

3. Interface Thermal Resistance in 2D Devices

3.1. Interface Thermal Resistance (ITR) between 2D Materials and Dielectric Contacts

The thermal conductance at interfaces is a critical consideration for heat management in the scaling down of modern nanoelectronics devices, at length scales that are comparable to the MFP of energy carriers. This interfacial thermal resistance, or thermal boundary resistance (Kapitzal resistance) was initially discovered for helium-solid system[122] and two theoretical models were accordingly developed, which are the acoustic mismatch model (AMM) and the diffusive mismatch model (DMM),[123] depending on whether the heat carriers (phonons) are specularly scattered (AMM) or randomly and elastically scattered (DMM) at the interface. Therefore, the interfacial thermal resistance could be influenced by many factors, such as the interface roughness, chemical bonding and the atomic defects.[124,125] For a more detailed theoretical modeling of heat transfer across interfacial boundaries, readers can refer to the review papers.[126,127]

For 2D based semiconductor devices, thermal transport at interfaces normally dominates in energy dissipation as well as in thermal management and thus understanding and characterization of interface heat transfer is necessary. Interface conditions also dominate the functionalities of these nanoelectronic devices. For example, trapped-charge induced Coulomb potential at the interface between the MoS_2 channel and the dielectric substrate is the dominant cause for the low carrier mobility in MoS_2 field-effect devices.[128] Just from a fundamental study point of view, 2D materials and especially their heterostructures provide an ideal platform to study interfacial heat transport along the cross-plane direction as a function of quantum-well exfoliation-controlled thickness. This has given rise to numerous studies in the past few years and this section is mainly focused on this topic. Modern advanced fabrication techniques have also enabled the realization of in-plane heterostructures[129,130] with atomically sharp interfaces. However, interfacial thermal resistance measurement of such small lateral interfaces is much more challenging compared to more popular interface measurements between different materials in the cross-plane direction.

Besides TDTR[131] and different 3ω techniques,[132] which are preferred for bulk interface measurement, noncontact Raman thermometry is an attractive technique to characterize atomic interface thermal transport, especially for 2D layered materials. The Raman-sensitive peaks of the measured sample would serve as a temperature thermometer while the incident heat source is from the heating by either electrical[133] or optical[134,135] means. A sketch of a Raman laser heating setup is shown in **Figure 7**a, where the interface thermal transport across monolayer MoS_2 and SiO_2/AlN substrate is measured.[136] For MoS_2, the positions of E^1_{2g} and A_{1g} show redshift with increasing temperature, which works as a temperature indicator, and its absorption power is obtained from the absorbed incident laser power of one free-standing monolayer MoS_2 by considering the substrate enhancement effect. Although MD simulations[124] show that the interface thermal conductance of MoS_2 on Au substrate is as high as 135–221 MW m^{-2} K^{-1}, the results obtained by the Raman technique are considerably lower–0.44 ± 0.07 MW m^{-2} K^{-1} for MoS_2-Au,[44] 1.94 MW m^{-2} K^{-1} for MoS_2-SiO_2,[59] 17.0 MW m^{-2} K^{-1} for MoS_2-h-BN,[137] and 15 MW m^{-2} K^{-1} for MoS_2-AlN.[136] The variations in the measured interface thermal conductance probably originate from the different interface quality prepared by the wet transfer method.[138] Higher values obtained are possibly due to the better interface quality in the absence of polymeric residues.[139] Temperature also affects the interface thermal transport as shown in Figure 7b, where the thermal boundary conductance (TBC) of MoS_2-SiO_2 exhibits a $T^{0.65}$ dependence. This temperature dependent interface thermal conductance agrees well with that of MoS_2-graphene interfaces simulated by MD calculation,[140] which is theoretically attributed to the higher temperature-induced strong scattering of inelastic phonons and the presence of excited phonons.

By changing the thickness of the channel and increasing the spacing between the sample and substrate, the interface thermal transport could be tuned significantly. Layer-dependent interface thermal conductance of MoS_2 with c-Si substrate was measured by means of micro-Raman spectroscopy, which increases monotonically with the number of layers.[138] Typically, the interface spacing leads to a higher Raman intensity and disturbs the local interface thermal transport as well. With decreasing number of layers, the Raman intensity difference between measured and theoretical values increases, implying a higher interface spacing, on the assumption that there is no spacing for the thickest MoS_2 (75 layers) for the theoretical Raman intensity calculation, which is shown in Figure 7c. The interface spacing would impair the interatomic forces between MoS_2 and substrates. Therefore, due to the higher interfacial energy coupling, thicker MoS_2 usually shows a better interface contact and the interface thermal conductance increases from 0.974 to 68.6 MW m^{-2} K^{-1} with increasing number of MoS_2 layers.[138] The thickness-dependence of interface thermal conductance is shown in Figure 7d. A similar trend of interface thermal conductance on thickness was shown between WSe_2 and SiO_2 in ref. [141]. The layer-controllable interface thermal conductance of 2D semiconductors is favorable for FET applications considering the improved heat dissipation to the substrates without degrading the channel electrical transport.

3.2. ITR between 2D Materials and Metal Contact

Electrical contacts to devices are usually made using metals. Thus the interfacial thermal resistance between metal and

沿 a 轴的温度和热导率 $^{-1\,-1}$ 为6.03W $^{-1\,-1}$ $_{-1}$， *沿* b 轴的值则低得多，为0.68W $^{-1\,-1}$ 。[107]

与过渡金属在过渡金属硫属化合物（TMDs）热导率中的作用类似，InSe的热导率计算值低于GaSe，这主要归因于声子速度的提升以及声光声子间散射的减少。[119–121] 此外，GaS的热导率总体上更高，这得益于其声子速度的进一步提升以及光声子模式与声子间耦合的减弱。

3. 二维器件的界面热阻

3.1. 二维材料与介电接触之间的界面热阻（ITR）

在现代纳米电子器件的微型化进程中，界面热导率是热管理的关键考量因素，其作用尺度与能量载体的平均自由程（MFP）相当。这种界面热阻（亦称热边界阻抗或卡皮察尔阻抗）最初是在氦-固体系统中被发现的 [122]，为此学者们提出了两种理论模型：声学失配模型（AMM）和扩散失配模型（DMM）。[123] 具体而言，前者假设声子在界面处发生镜面散射，后者则认为其呈现随机弹性散射。界面热阻受多种因素影响，包括表面粗糙度、化学键合强度及原子缺陷等。[124,125] 若需更深入的界面传热理论建模，读者可参考相关综述论文。[126 ， 127]

对于基于二维材料的半导体器件而言，界面热传导通常主导着能量耗散以及热管理过程，因此理解和表征界面热传递是必要的。界面条件也主导着这些纳米电子器件的功能特性。例如，MoS_2 通道与介电衬底界面处由俘获电荷诱导产生的库仑势，是MoS_2 场效应器件载流子迁移率低的主要原因。[128] 仅从基础研究的角度来看，二维材料及其异质结构为研究量子阱剥离控制厚度方向的面内界面热传导提供了理想平台。这在过去几年引发了大量研究，本节主要聚焦于这一主题。现代先进制造技术还实现了具有原子级锐利界面的面内异质结构 [129,130] 。然而，相较于在材料间横向进行的常规界面测量，这类微小横向界面的热阻测量难度要大得多。

除了 TDTR [131] 和不同的3 ω 技术 [132]（这些技术主要用于块体界面测量）之外，非接触式拉曼测温技术是一种很有吸引力的技术，可用于表征原子界面热传输，特别是二维层状材料。测量样品的拉曼敏感峰可作为温度计，而入射热源则来自电加热 [133] 或 光学 [134,135] 一种 拉曼激光加热装置的示意图如 **图7a** 所示，其中测量了单层MoS_2 和SiO_2 /AlN基底之间的界面热传输。[136] 对于MoS_2，$E^1{}_{2g}$ 和A_{1g} 的位置随温度升高而红移，这可作为温度指示器，其吸收功率通过考虑基底增强效应，从单层MoS_2 的入射激光功率吸收量中获得。尽管分子动力学模拟 [124] 显示MoS_2 在金基底上的界面热导率高达135–221MWm $^{-2}$ K $^{-1}$，但 theresultsobtainedbythe-Ramantechnique 的测量值存在显著的 lower−0.44 ±0偏差。07 MW m $^{-2}$ K $^{-1}$ 对于MoS_2 -Au，[44]
1.94 MoS_2 -SiO_2 的热导率MW $^{-2}$ K $^{-1}$ 为17.0 MW $^{-2}$ K $^{-1}$，MoS_2 -h-BN为15 MW m $^{-2}$ K $^{-1}$，MoS_2-AlN 为15 MW [136] m $_{-2}$ K $^{-1}$。测量的界面热导率变化可能源于湿法转移法制备的界面质量不同。[138] 较高的数值可能是由于无聚合物残留的界面质量更好。如图7b所示，温度也会影响界面热传输，其中MoS_2 -SiO_2 的热边界导热系数（TBC）表现出 0.65 的温度依赖性。这种温度依赖的界面热导率与分子动力学模拟的MoS_2 -石墨烯界面的热导率吻合良好，[140] 这在理论上归因于温度诱导的非弹性声子强散射和激发声子的存在。

通过改变通道厚度并增加样品与基底之间的间距，界面热传输可以显著调节。利用微拉曼光谱测量了MoS_2 与c-Si基底的层依赖性界面热导率，其随层数单调增加。[138] 通常，界面间距会导致更高的拉曼强度，并干扰局部界面热传输。随着层数减少，测量值与理论值之间的拉曼强度差异增大，这意味着界面间距更大——假设在理论拉曼强度计算中，最厚的MoS_2（75层）没有间距，如图7c所示。界面间距会削弱MoS_2 与基底之间的原子间作用力。因此，由于更高的界面能耦合，较厚的MoS_2 通常表现出更好的界面接触，界面热导率从0.974增加到68。6 MW m $^{-2}$ K $^{-1}$ 随着MoS_2 层数量的增加。图7d展示了界面热导率随厚度变化的趋势。参考文献[141]中也展示了WSe_2 与SiO_2 之间界面热导率随厚度变化的类似趋势。二维半导体的层可控界面热导率对于场效应晶体管（FET）应用具有优势，因为其在不降低沟道电输运性能的前提下，能有效改善向衬底的散热。

3.2. 二维材料与金属接触的ITR

设备的电气接触通常采用金属材料。因此，金属与

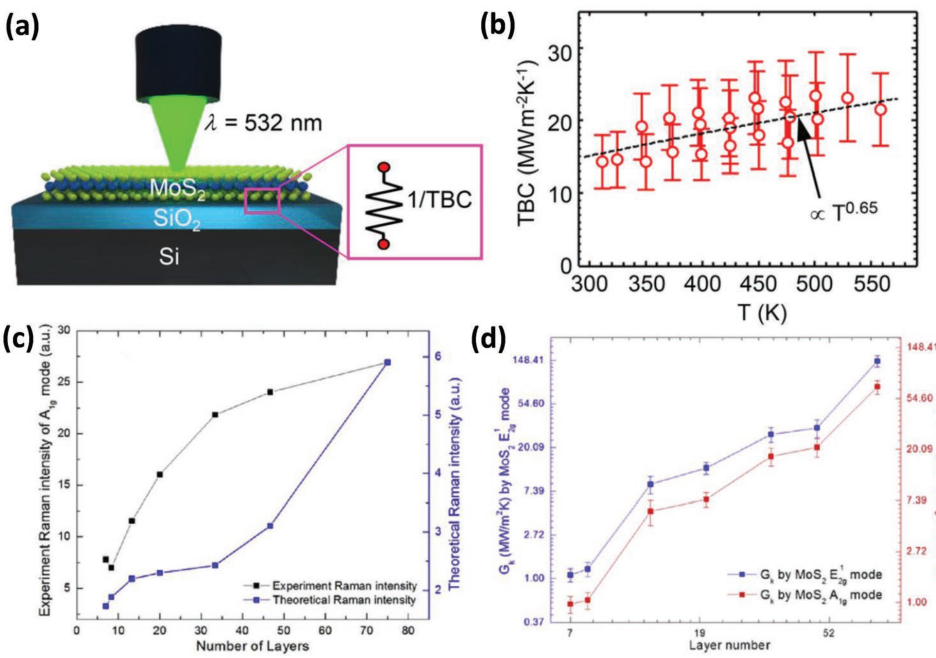

Figure 7. a) Sketch of monolayer MoS_2 on substrate and Raman thermometry measurement setup. b) Temperature dependent thermal boundary conductance (TBC) across MoS_2-SiO_2, which follows trend of $T^{0.65}$. The graphs are from paper. Reproduced with permission.[136] Copyright 2017, American Chemical Society. c) Change of measured- and theoretical calculated- Raman peak intensity of A_{1g} mode for MoS_2 samples. d) Layer-dependent interface thermal conductance of MoS_2-c-Si deduced from Raman sensitive $E_{2g}{}^1$ and A_{1g} modes. Reproduced with permission.[138] Copyright 2017, Elsevier.

2D materials plays a critical role in thermal management of future 2D devices. There are two types of interface geometry: side contact and edge contact. Side contact can be made by contacting the metal surface with the plane of 2D materials. However, large contact resistance, for both electron and phonon, usually exists at the interface which drastically restrains the device current.[142] Mao et al. studied interfacial thermal resistance in the side contact configuration between MoS_2 and Sc (Ru) through chemical bonding, and that between MoS_2 and Au (Pd) via weak physical bonding.[143] As expected, the physisorbed case with a weaker bonding results in a much larger interfacial thermal resistance than the chemisorbed case. Compared with graphene/metal chemisorbed system, the metal/MoS_2 chemisorbed contact is significantly more resistive, with almost 10-times higher thermal resistance. Therefore, for MoS_2 devices with side contact to metal electrodes, the interfacial thermal resistance is an impediment to heat dissipation.

On the other hand, for edge contacts, strong overlapping of electronic orbitals between the 2D material and metal electrode in most cases can lead to a substantial reduction in electron tunnel barrier and a great increase in electrical transport, which has been experimentally reported for both graphene[144] and MoS_2 devices.[145] By using molecular dynamics simulations, Liu et al. studied the interfacial thermal transport across the monolayer MoS_2 and Au electrode, where an edge contact was formed.[124] The effect of surface orientation of the crystal gold electrode is also considered. As shown in **Figure 8**, although there is a remarkable dependence of interfacial thermal conductance on the interfacial structure, overall, the interfacial thermal conductance is large. For example, the largest value of interfacial thermal conductance occurs when $\theta = 0$ at the (110) surface, about 2.21×10^8 W m^{-2} K^{-1}. This is comparable to many commonly studied contact in semiconductor devices, for example, the Si/Ge (3×10^8 W m^{-2} K^{-1}),[146] Au–Si (1.88×10^8 W m^{-2} K^{-1}),[147] Al–Si (4.5×10^8 W m^{-2} K^{-1}),[148] and Cu–Si (4×10^8 W m^{-2} K^{-1})[149] interfaces. It is worth emphasizing that although the thermal conductivity of graphene is nearly two orders higher than that of monolayer MoS_2, the contact thermal conductance of monolayer MoS_2 with metal contact is comparable to that of metal–graphene interfaces with a chemically bonded structure (2.5×10^8 W m^{-2} K^{-1}).[150] Thus edge contact between MoS_2 and metal electrodes has advantages in thermal management, in addition to the larger electrical current, compared to the side contact configuration.

In addition, the interfacial thermal conductance also depends on the interface configuration. At the MoS_2/Au (001) contact, the interfacial thermal conductance is robust and weakly dependent of the orientation angle. On the other side, for both MoS_2/Au (111) and MoS_2/Au (110) contacts, the interfacial thermal conductance decreases markedly when θ increases from 0° to 30°. For all the considered contacts, the interfacial thermal conductance decreases with the number of S vacancies, following a linear dependence.

4. Phonon-Driven Emerging Applications of 2D Semiconductors

4.1. Thermoelectric

The thermoelectric effect describes the conversion between heat and electrical energy.[29,151,152] Heat is directly converted

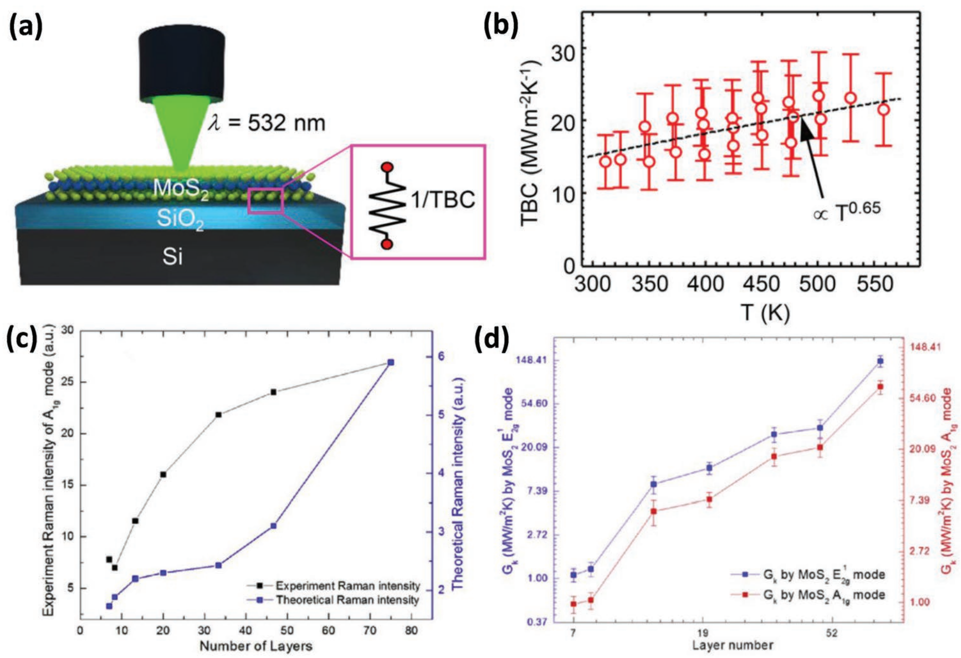

图7. a) 基底上单层MoS_2的示意图及拉曼热测量装置。b) MoS_2-SiO_2跨MoS的温度依赖性热边界导热率（TBC），其趋势与$T^{0.65}$相符。图表引自论文，经许可转载。[136] 版权2017年美国化学会。c) MoS_2样品A_{1g}模式拉曼峰强度的实测与理论计算变化。d) 通过拉曼敏感的$E_{2g}{}^1$和A_{1g}模式推导出的MoS_2-c-Si界面热导率随层厚度的变化。经许可转载。[138] 版权2017年爱思唯尔。

二维材料在未来的二维器件热管理中起着关键作用。界面几何结构有两种类型：侧接触和边缘接触。侧接触可以通过将金属表面与二维材料平面接触来实现。然而，电子和声子在界面处通常存在较大的接触电阻，这极大地限制了器件电流。[142] Mao等人研究了通过化学键合在MoS_2和Sc（Ru）之间，以及通过弱物理键合在MoS_2和Au（Pd）之间侧接触配置的界面热阻。[143] 如预期，物理吸附情况下的弱键合导致界面热阻远大于化学吸附情况。与石墨烯/金属化学吸附系统相比，金属/$_2$MoS化学吸附接触的电阻显著更高，热阻几乎高出10倍。因此，对于MoS_2与金属电极采用侧接触的器件，界面热阻是散热的障碍。

另一方面，对于边缘接触，二维材料与金属电极之间的电子轨道在大多数情况下存在强烈的重叠，这会导致电子隧穿势垒显著降低和电输运大幅增加，这一现象已在石墨烯[144] 和MoS_2器件的实验中得到报道。[145] 通过分子动力学模拟，Liu等人研究了单层MoS_2和金电极之间形成边缘接触时的界面热输运，并考虑了金电极晶体表面取向的影响。如 **图8** 所示，尽管界面热导率对界面结构具有显著依赖性，但总体而言，界面热导率仍然较大。例如，当θ=0时，界面热导率达到最大值。

(110) 表面热导率约为2.21×10^8 W m^{-2} K^{-1}。这与半导体器件中常见的多种接触界面热导率相当，例如Si/Ge（3×10^8 W m^{-2} K^{-1}）、[146] Au–Si（1.88×10^8 W m^{-2} K^{-1}）、[147] Al–Si（4.5×10^8 W m^{-2} K^{-1}）、[148] 和Cu–Si（4×10^8 W m^{-2} K^{-1}）[149] 界面。需要强调的是，尽管石墨烯的热导率比单层MoS高出近两个数量级$_2$，但单层MoS_2与金属接触的热导率与化学键合结构的金属–石墨烯界面热导率相当（2.5×10^8 W m^{-2} K^{-1}）。[150] 因此，与侧接触配置相比，MoS_2与金属电极之间的边缘接触除了具有更大的电流外，在热管理方面也具有优势。

此外，界面热导还取决于界面构型。 在MoS_2/Au（001）接触中，界面热导稳定且对取向角的依赖性较弱。另一方面，对于MoS_2/Au（111）和MoS_2/Au（110）接触，当θ从0°增加到30°时，界面热导显著降低。对于所有考虑的接触，界面热导随S空位数量的增加而减少，呈现线性依赖关系。

4. 声子驱动的二维半导体新兴应用

4.1. 热电

热电效应描述了热能与电能之间的转换。[29,151,152] 热能直接转换为电能

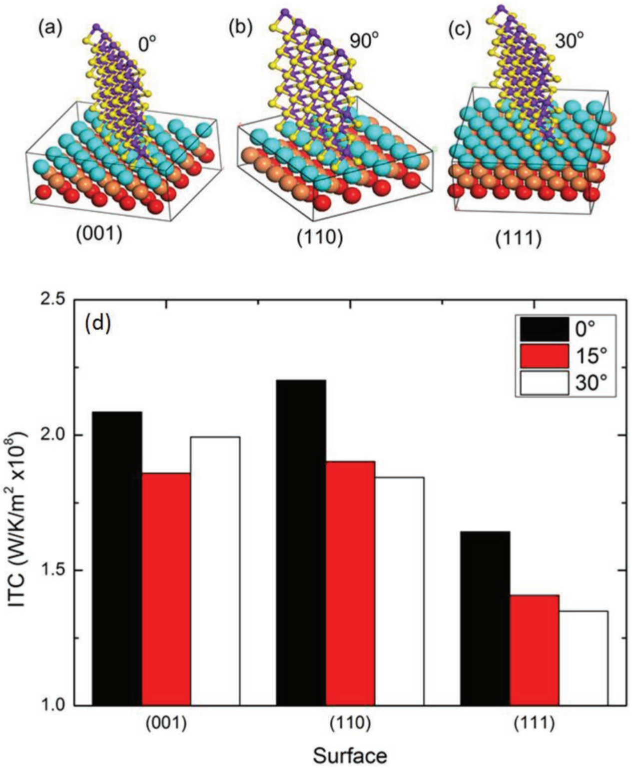

Figure 8. a–c) MoS_2 on (001), (110), and (111) Au surfaces, respectively. Mo and S atoms are shown in purple and yellow color, and Au atoms in the first, second, and third layers are shown in cyan, brown, and red color, respectively. d) Interfacial thermal conductance on Au crystal surfaces with different contacting cases. Reproduced with permission.[124] Copyright 2016, Springer.

into electricity through the Seebeck effect, and an electric current flow would lead to heat absorption and release, which is known as Peltier effect.[153] The reversible generation of heat flow when an electric current passes through a conductor with a temperature gradient is called the Thomson effect.[154] The efficiency of TE devices is normally characterized by a dimensionless figure of merit $ZT = (S^2T)/(\rho\kappa)$, where S is the Seebeck coefficient, T is the temperature, ρ is the electrical resistivity, and κ is the thermal conductivity, including both electron (κ_e) contributions and phonon (κ_L) contributions. Thus, a higher ZT factor follows from a low thermal conductivity to generate a large temperature gradient, and high electrical conductivity to conduct electricity efficiently–desirable TE materials are those that behave like a phonon-glass electron-crystal.[29,155] Typically, optimizing the material parameters is an exercise in compromise since they are correlated to each other. For example, the electrical conductivity of a material increases with increasing carrier concentration while the Seebeck coefficient decreases as it is physically related to the average entropy per charge carrier. Meanwhile, the carrier concentration induces a larger electrical thermal conductivity as related by the Wiedemann–Franz law.[156,157] Therefore, good TE materials are typically degenerate semiconductors, as shown in **Figure 9**. According to Hicks and Dresselhaus,[25,158] low dimensional nanostructures perform favorably for TE applications due to quantum confinement. Considering the staircase-like energy dependent density of states (DOS) of 2D materials shown in Figure 9b, 2D materials show better TE properties than those of bulk ones.

By electrolyte gating method, Pu et al. measured carrier density-dependent TE properties of monolayer MoS_2 and WSe_2.[159] In this work, the authors fabricated the electric double-layer transistors based on centimeter-scale MoS_2, $MoSe_2$, and WSe_2 monolayers. In electric double-layer transistors, the dielectric layers of transistors are replaced with electrolytes. Therefore, it is possible to continuously increase the carrier density up to 5×10^{13} cm^{-2} and realize precise control of the Fermi level and Seebeck coefficient, owing to the high specific capacitance of electric double-layer. As shown in **Figure 10**, the absolute value of Seebeck coefficient of the monolayer samples is comparable with that of the bulk samples. However, at a similar Seebeck coefficient, the electrical conductivity of the 2D monolayers is a few orders of magnitude higher than that of the bulk samples. It turns out that the power factor of monolayer MoS_2 and WSe_2 is nearly one order of magnitude higher than that of their bulk materials counterparts, which provides direct evidence for the advantage of 2D monolayer sheets in thermoelectric application. Figure 10a,b compares the Seebeck coefficient and power factor of monolayers and their 3D bulk counterparts.

Given the combination of field-effect doping that can be well controlled, atomically clean surfaces, and quantum-well thickness variation, 2D semiconductors provide a versatile platform

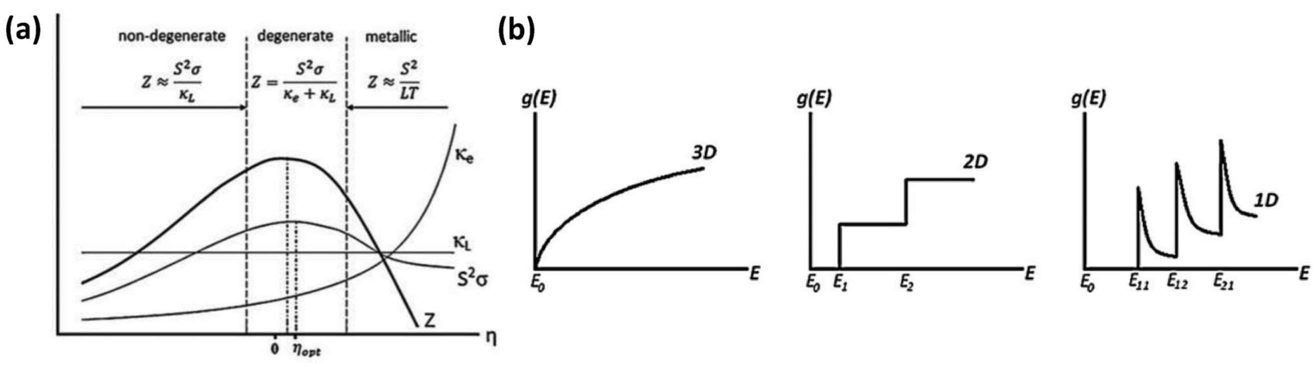

Figure 9. a) Figure of merit as a function of reduced Fermi level η[151] and η is defined as $\eta = (E_F - E_0)/k_BT$, where E_F is the Fermi energy, k_B is the Boltzmann constant, and T is temperature. b) DOS, $g(E)$, for bulk materials (3D), quantum well (2D), and quantum wire (1D). 2D is favorable for high power factor due to its staggered DOS. Reproduced with permission.[151] Copyright 2010, Elsevier.

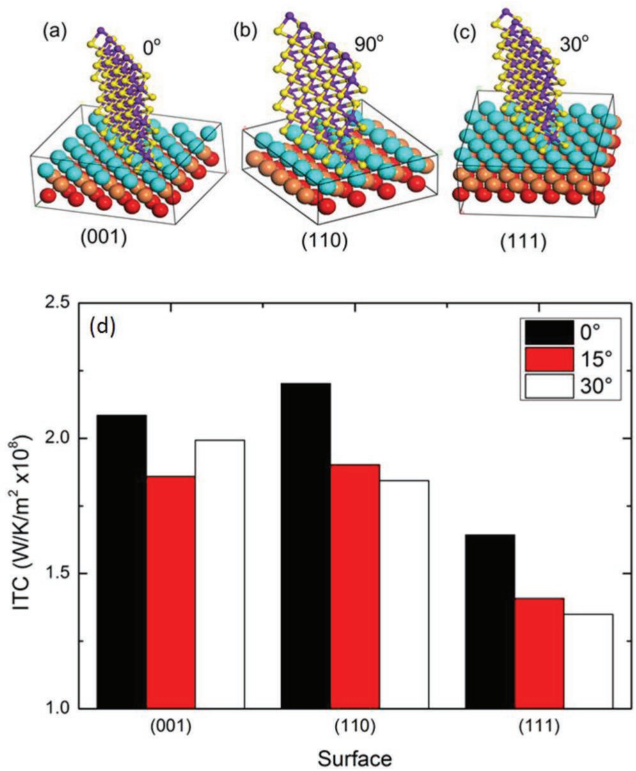

图8. a–c) MoS_2 在（001）、（110）和（111）金表面的分布情况。钼和硫原子分别用紫色和黄色表示，第一、第二和第三层的金原子分别用青色、棕色和红色表示。d) 不同接触情况下金晶体表面的界面热导率。经许可转载。[124] 版权所有2016年，Springer。

通过塞贝克效应将能量转化为电能，而电流流动则会产生热吸收与释放现象，这一现象被称为珀尔帖效应。当电流流经存在温度梯度的导体时，[153] 会产生可逆的热流，这种现象称为汤姆逊效应。[154] 热电装置的效率通常用无量纲性能指标 $ZT=(S^2T)/(\rho\kappa)$ 来表征，其中 S 为塞贝克系数，T 表示温度，ρ 为电阻率，κ 则是包含电子热导率（κ_e）在内的热导率。热电效应主要由载流子贡献和声子（κ_L）贡献共同决定。要获得更高的 ZT 因子，需要具备低热导率以形成显著的温度梯度，同时保持高电导率以实现高效导电——理想的热电材料应兼具声子玻璃、电子和晶体的特性。[29,155] 通常，材料参数的优化需要权衡取舍，因为这些参数之间存在相互关联。例如，材料的电导率会随着载流子浓度的增加而提升，但塞贝克系数会相应降低，这与其物理特性——即每个载流子的平均熵值密切相关。与此同时，载流子浓度还能通过维德曼-弗兰兹定律增强电热导率。[156,157] 因此，优质的热电材料通常属于简并半导体，如 **图9** 所示。根据希克斯和德雷塞尔豪斯的研究，[25,158] 低维纳米结构由于量子限制效应，在热电应用中展现出显著优势。如图9b所示，二维材料具有阶梯状的能依赖态密度（DOS），因此其热电性能优于块体材料。

通过电解质门控法，Pu等人测量了单层MoS_2和WSe_2的载流子密度依赖性热电特性。[159] 在本研究中，作者基于厘米级MoS_2、$MoSe_2$和WSe_2单层制备了双电层晶体管。在双电层晶体管中，晶体管的介电层被电解质取代。因此，由于双电层具有高比电容特性，可以将载流子密度连续提升至$5\times10^{13}\ cm^{-2}$，并实现费米能级和塞贝克系数的精确控制。如 **图10** 所示，单层样品的塞贝克系数绝对值与块体样品相当。然而，在相似的塞贝克系数下，二维单层的电导率比块体样品高出几个数量级。研究发现，单层MoS_2和WSe_2的功率因子比其块体材料对应物高出近一个数量级，这为二维单层片在热电应用中的优势提供了直接证据。图10a，b比较了单层材料及其三维块体对应物的塞贝克系数和功率因子。

二维半导体凭借其可控的场效应掺杂、原子级洁净表面及量子阱厚度可调等优势，成为多功能材料平台。

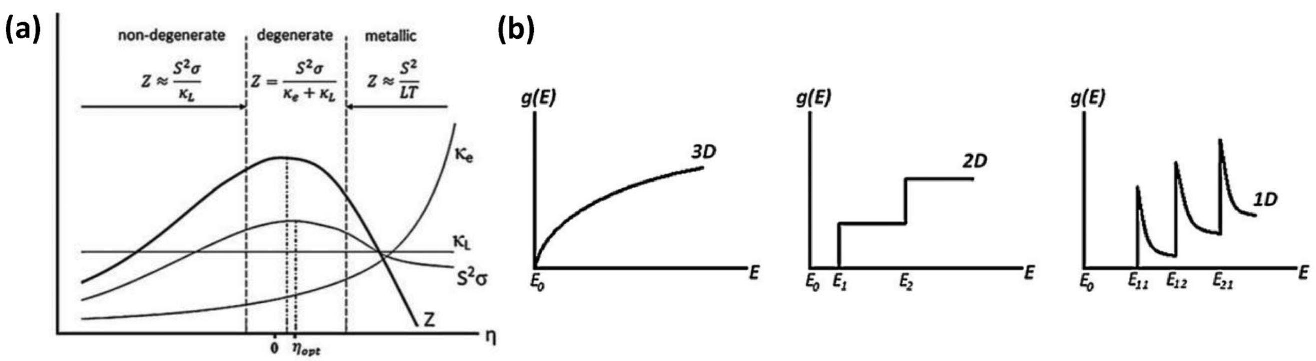

图9. a) 优值 随费米能级 [151] 减量变化的函数关系图。η 定义为 $\eta=(E_F-E_0)/k_BT$，其中 E_F 为费米能级，k_B 为玻尔兹曼常数，T 为温度。b)体材料（三维）、量子阱（二维）和量子线（一维）的 *功率因子密度* g (E)。二维材料由于具有交错的功耗密度，能获得更高的功率因子。经授权转载。[151] 版权所有2010年爱思唯尔。

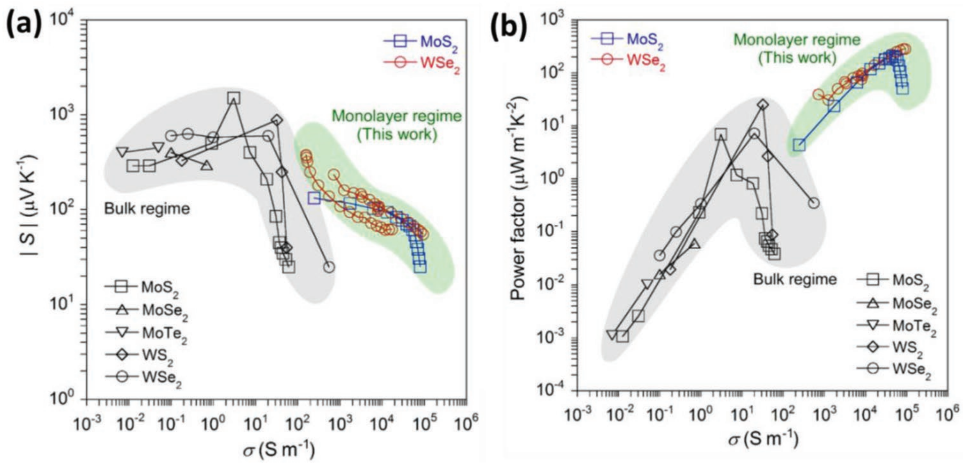

Figure 10. Comparison of electrical conductivity-dependent Seebeck coefficient a) and power factor b) for various TMDs, including bulk and monolayer MoS_2, $MoSe_2$, $MoTe_2$, WS_2, and WSe_2. Reproduced with permission.[159] Copyright 2016, American Physical Society.

to study the TE effect. MoS_2, as a representative 2D semiconductor, has shown attractive thermal as well as thermoelectric properties. Numerous theoretical papers[160,161] have predicted its superior TE properties while spin-dependent thermal transport has been studied for sandwiched S–Mo–Se structure (Janus SMoSe).[162] Earlier TE measurements focused on its photo-thermoelectric effect[163] and the measured Seebeck coefficient value of single layer MoS_2 was between -4×10^2 to -1×10^5 μV K^{-1}. Wu et al.[164] later measured the Seebeck coefficient of monolayer CVD grown MoS_2 with values ≈30 mV K^{-1}, even though the conductivity was low and the authors demonstrated that the transport was dominated by variable-range hopping (VRH). Kayyalha et al.[165] reported the thermoelectric transport properties of MoS_2 with different number of layers and it was shown that both electrical conductivity and Seebeck coefficient demonstrate a strong dependence on thickness (**Figure 11**a,b). With reducing thickness, the thermoelectric power factor (PF) keeps increasing until a peak, where a high ≈50 μW cm^{-1} K-2 PF was shown for two-layer MoS_2 sample in its ON state, before dropping greatly for the monolayer, which was theoretically attributed to a difference in the energy dependence of the electron MFP. A similar thickness-dependent PF trend was reported in paper,[166] where a relatively higher PF of two-layer MoS_2 was demonstrated due to the better conductivity in the degenerate metallic regime and a large Seebeck coefficient due to the high valley degeneracies and large effective masses. Quite recently, Wu et al.[167] showed that the PF of layered MoS_2 can be even enhanced by a strong interaction of electrons at the Fermi level with a local magnetic impurity (i.e., sulfur vacancy) and this so-called magnetic impurity-induced Kondo effect leads to a new record PF around 50 mW m^{-1} K^{-2}, as shown in Figure 11d. The single-atom sulfur vacancy in naturally grown MoS_2, confirmed by low-temperature scanning tunneling microscopy (STM) induces magnetic states, which further leads to a significant band splitting of ≈50 ± 5 meV at the conduction subband adjacent to the sulfur vacancy. The Kondo effect induced by magnetic impurities is seen to be an effective way to tune the Seebeck coefficient and *ZT* value for 2D semiconductor materials. Different from n-type MoS_2, WSe_2 shows interesting p-typed transport behavior. By means of gate-controlled surface carrier doping, Yoshida et al.[168] showed that the optimization of power factor is possible in WSe_2 single crystals and the optimized values are comparable to those of a popular TE material, bismuth telluride, which is measured as 37 and 32 μW cm^{-1} K^{-2} for p- and n- type conduction transport, respectively.

Like TMDs that have high mobility and tunable bandgap, layered black phosphorus has also been studied for its TE properties. As discussed in Section 2.2, BP exhibits unique anisotropic transport for both electrons and phonons, which is beneficial for TE performance, since the high electrical and thermal conductance directions are orthogonal to each other, enabling values of *ZT* above 1 for monolayer BP at a doping density of $\approx 2 \times 10^{20}$ cm^{-2} along the AC orientation.[169] Later experimental work was carried out on thicker material and it was found that Seebeck coefficient could reach up to +510 μV K^{-1} in the hole depleted state at 210 K by using an EDLT configuration.[66] Thickness-dependent thermoelectric characteristics of thick BP was studied as well,[170] where Mott's VRH model was demonstrated to be the dominant mechanism in the TE and electrical transport of BP,[170,171] as for the case of MoS_2.

Other 2D semiconducting materials, like n-type TiS_2[172] and InSe[173] have been recently explored as well for TE applications. As mentioned earlier, due to their weak interlayer van der Waals force, 2D layered semiconductors can be mechanically exfoliated into well-defined thickness down to single layer of a few angstroms thickness.[174,175] By measuring thickness dependent TE properties in InSe, Zeng et al.[176] have recently shown that the power factor increases greatly due to the sharper edge of the conduction-band DOS caused by quantum confinement and a Seebeck coefficient up to 570 μV K^{-1} in 7 nm InSe sample has been reported. Further analysis shows that power factor in InSe would be increased significantly when the confined dimension is smaller than the thermal de Broglie wavelength, a scenario estimated by theoretical calculation.[177] When the confined length increases beyond than the thermal de Broglie wavelength, the power factor decreases gradually toward the value for bulk InSe,[178] which is shown in **Figure 12**.

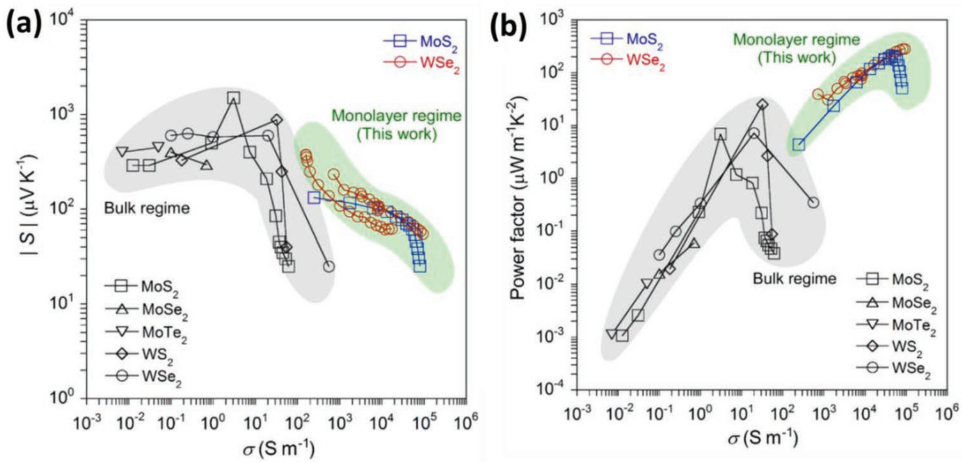

图10. 不同过渡金属硫属化合物（TMDs）的电导率依赖性塞贝克系数a)和功率因数b)对比，包括块体和单层MoS_2、$MoSe_2$、$MoTe_2$、WS_2和WSe_2。经许可转载。[159] 版权所有2016年美国物理学会。

研究热电效应。 MoS_2作为典型的二维半导体，展现出优异的热学及热电性能。大量理论论文[160,161]预测其具有卓越的热电特性，而自旋依赖热输运现象已在夹层S–Mo–Se结构（Janus SMoSe）中得到研究。[162]早期热电测量主要关注其光热电效应[163]，单层MoS的测得塞贝克系数值$_2$介于-4×10^2至-1×10^5 μV K^{-1}之间。Wu等人[164]后续测量了单层CVD生长的MoS_2的塞贝克系数，其值约为30 mV K^{-1}，尽管导电性较低，且作者证实输运主要由可变范围跳跃（VRH）主导。Kayyalha等人[165]报道了不同层数的MoS_2的热电输运特性，并表明其电导率和塞贝克系数均对厚度表现出强烈依赖性（**图11** a，b）。随着厚度减小，热电功率因子（PF）持续增加直至达到峰值，其中双层MoS_2样品在导通状态下测得高值≈50 μW cm^{-1} K－2 PF，而单层样品则大幅下降，这在理论上归因于电子平均自由程（MFP）能量依赖性的差异。类似厚度依赖的PF趋势在文献[166]中也有报道，其中双层MoS_2因退化金属态下的优异导电性以及高谷简并度和大有效质量导致的较大塞贝克系数，展现出相对更高的PF。最近，Wu等人[167]的研究表明，层状MoS_2的泡浦效应（PF）甚至可以通过费米能级处电子与局域磁性杂质（即硫空位）的强相互作用得到增强，这种所谓的磁性杂质诱导的Kondo效应导致了约50 mW m^{-1} K^{-2}的新纪录泡浦效应，如图11d所示。天然生长的MoS_2中单原子硫空位通过低温扫描隧道显微镜（STM）确认，会诱导磁态，进而导致硫空位相邻的导电子带出现约50±5 meV的显著带分裂。磁性杂质诱导的Kondo效应被证明是调节二维半导体塞贝克系数和 *ZT* 值的有效方法。

材料。与n型MoS_2不同，WSe_2展现出有趣的p型传输行为。通过栅控表面载流子掺杂，Yoshida等人[168]表明在WSe_2单晶中可以优化功率因子，且优化值与流行的TE材料碲化铋相当，其p型和n型传导传输的测量值分别为37和32 μW cm^{-1} K^{-2}。

与具有高迁移率和可调带隙的过渡金属硫化物（TMDs）类似，层状黑磷也因其热电特性而受到研究。如第2.2节所述，BP对电子和声子都表现出独特的各向异性输运特性，这对热电性能有利，因为其高电导率和热导率方向彼此正交，使得单层BP在掺杂密度约为2×10^{20} cm^{-2}时沿AC取向的 *ZT* 值可超过1。[169]后续实验研究在更厚材料上进行，发现采用EDLT结构时，空穴耗尽态下的塞贝克系数可在210 K时达到+510 μV K^{-1}。[66]对厚黑磷的厚度依赖性热电特性也进行了研究，[170]其中Mott的VRH模型被证明是黑磷热电和电输运的主要机制，[170,171]与MoS_2的情况类似。

其他二维半导体材料，如n型TiS_2[172]和InSe[173]近年来也被探索用于热电应用。如前所述，由于其层间范德华力较弱，二维层状半导体可被机械剥离成厚度明确的单层结构，厚度可达数埃。[174,175]通过测量InSe的厚度依赖性热电特性，曾等人[176]最近表明，由于量子限制导致的导带态密度边缘更尖锐，功率因数显著增加，且7纳米InSe样品的塞贝克系数高达570 μV K^{-1}。进一步分析表明，当受限尺寸小于热德布罗意波长时（理论计算估计），InSe的功率因数将显著增加。[177]当受限长度超过热德布罗意波长时，功率因数逐渐降低至块体InSe的值，[178]如 **图12** 所示。

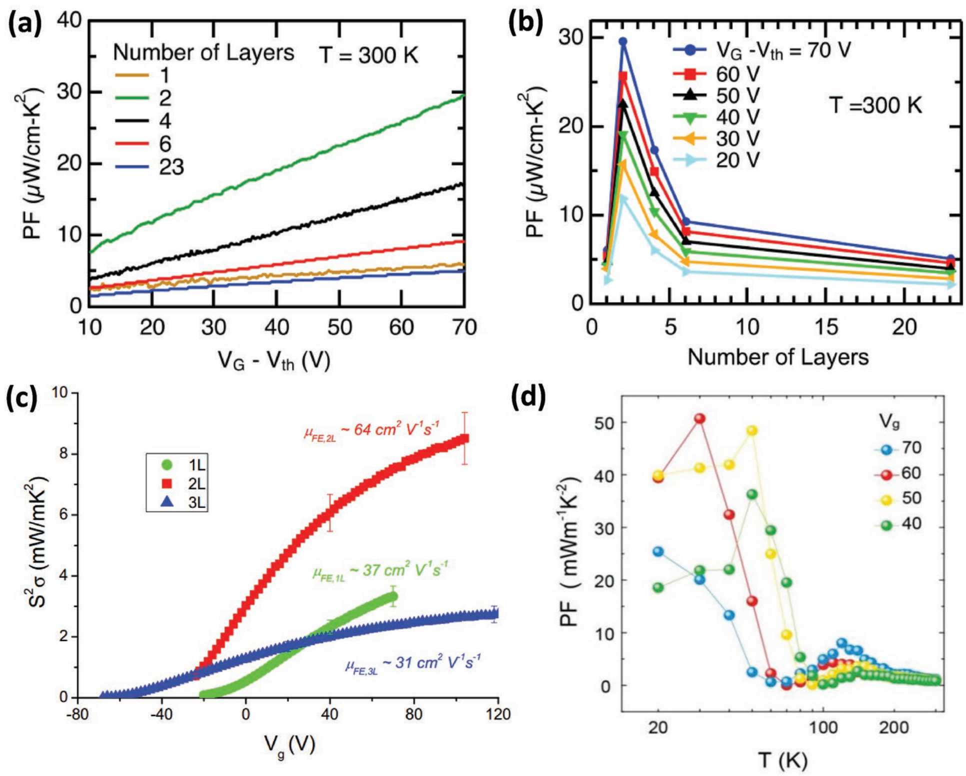

Figure 11. a) Back-gate voltage (V_g-V_{th}) dependent thermoelectric power factor (PF) for different layers of MoS_2. b) PF versus the number of layers at different V_g-V_{th}. All the PF values are for room temperature. Reproduced with permission.[165] Copyright 2016, American Institute of Physics. c) PF as a function of back-gate voltage. The bilayer MoS_2 shows a maximum PF with a larger electrical mobility. Reproduced with permission.[166] Copyright 2017, American Physical Society. d) PF versus measured temperature for different back-gate voltages. An abnormal increase of PF is exhibited due to the Kondo effect. Reproduced with permission.[167] Copyright 2019, arXiv.

4.2. Photoelectric (Electron–Phonon Coupling)

In addition to the direct effect to heat-to-electric energy conversion, phonons also play important roles in other energy conversion mechanisms, such as the photoelectric effect. Light–matter interaction is highly sensitive to the thermal environment of the material during the measurement. The coupling of the phonon and electron can renormalize the electronic structure, which can affect the light emission and absorption. Photocarriers, electrons and holes, can be scattered through a series of phonon coupling mechanisms satisfying the selection rules subject to energy and momentum conservation. In addition, thermal vibration distorts the ideal atomic positions and scatters the light-triggered electron–hole pairs. Specifically, acoustic phonons are involved in the carrier relaxation process via deformation potential interaction[179] while optical/homopolar phonons are through Fröhlich interaction. For instance, TMDs such as MoS_2 and WS_2 show a strong absorption of visible light due to the "band nesting" effect. Photocarriers produced in the band-nesting part (located midway between the Γ and Λ points) drifts toward their immediate band extrema: Λ valley and Γ hill for electrons and holes, respectively.[180] These electron–hole pairs are separated in the k-space and their radiative recombination requires either emission or absorption of single or multiple phonons. This indirect emission process results in low yield and the carrier lifetime can be estimated to be 1 ns,[181] which is in contrast to direct excitons with lifetime of around 100 ps and a higher quantum yield.[182] Similarly, by using femtosecond time-resolved photoemission electron microscopy, the carrier dynamics property of monolayer WSe_2 on SiO_2/Si substrate has been studied to uncover the role of the intervalley electron–phonon scattering of electrons.[183]

In monolayer or few-layer TMDs energy dissipation during nonradiative emission plays a vital role in the ultrafast processes of the carriers. Generally, two pathways of nonradiative energy channels exist after photoexcitation of layered TMDs[184] (see **Figure 13**): 1) ultrafast cooling of hot carriers via electron–phonon scattering and subsequent formation of excitons, and 2) nonradiative recombination of excitons at the surface. While thermalization of the MoS_2 leads to a redshift of the exciton resonance energy due to electron–phonon scattering, the excess energy released from the cooling of hot carriers can be dissipated and transferred to the phonons within ≈2 ps for pathway 1), while process 2) is a Shockley–Read–Hall recombination responsible for energy dissipation from surfaces to external phonons within ≈9 ps.

In addition, phonons are also found to be important for the production of the excitons, a type of quasiparticle playing a significant role in the optical properties of layered semiconducting materials. The strength of the exciton-phonon can be obtained by experimentally measuring the dephasing

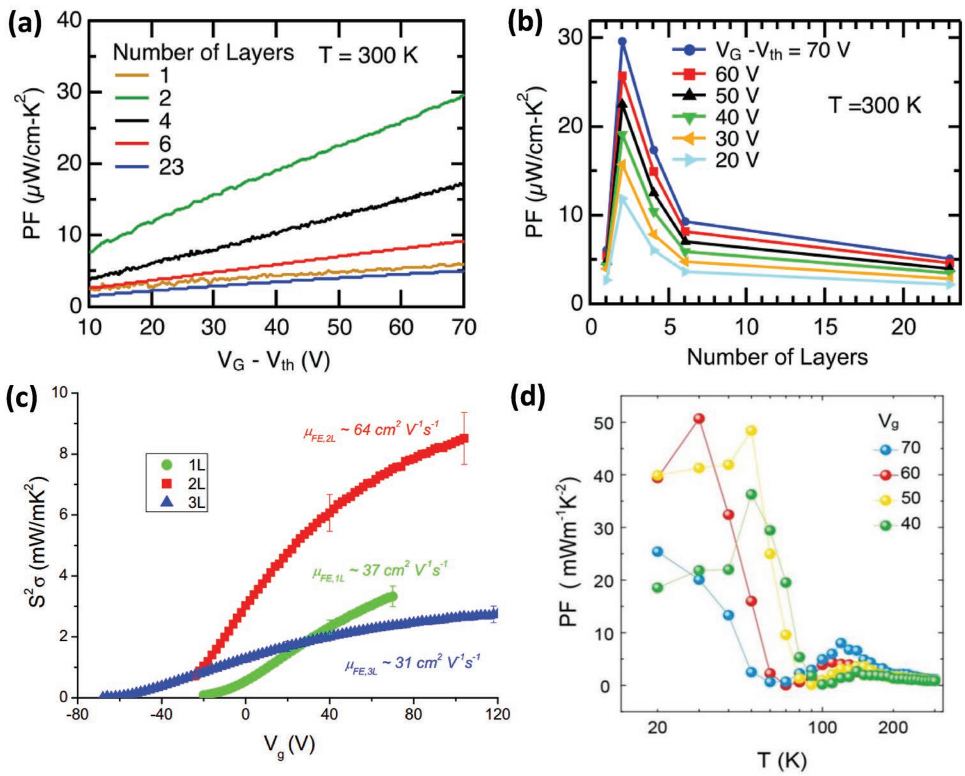

图11. a) 不同MoS *2* 层的背栅电压（*V* g-*V* th）依赖热电功率因子（PF）2。b) PF与不同 *V* g-*V* th 下层数量的关系。所有PF值均为室温数据。经授权转载。[165] 版权2016年美国物理学会。c) PF随背栅电压变化的函数关系。双层MoS 2 展现出最大PF值，同时具有更高的电迁移率。经授权转载。[166] 版权2017年美国物理学会。d) 不同背栅电压下PF与测量温度的关系。由于Kondo效应，PF呈现异常增长。经授权转载。[167] 版权2019年arXiv。

4.2. 光电效应（电子-声子耦合）

除了对热能到电能转换的直接影响外，声子在其他能量转换机制中也发挥着重要作用，例如光电效应。光-物质相互作用对材料在测量过程中的热环境高度敏感。声子与电子的耦合可以重整化电子结构，从而影响光的发射和吸收。光载流子（电子和空穴）可以通过一系列满足能量和动量守恒选择规则的声子耦合机制发生散射。此外，热振动会扭曲理想原子位置并散射光触发的电子-空穴对。具体而言，声学声子通过形变势相互作用参与载流子弛豫过程 [179]，而光学/同极声子则通过弗罗利希相互作用参与。例如，过渡金属硫属化合物（如MoS 2 和WS 2）由于“能带嵌套”效应而表现出强烈的可见光吸收。在能带嵌套区（位于Γ点与^点之间的中点）产生的光生载流子，会向其附近的能带极值漂移——电子向^谷移动，空穴则向Γ峰移动。[180] 这些电子-空穴对在 *k* 空间中被分离，其辐射复合过程需要发射或吸收单个或多个声子。这种间接辐射机制导致载流子生成效率较低，且载流子寿命较短。

其寿命可估算为1纳秒，[181] 这与寿命约100皮秒且量子产率更高的直接激子形成鲜明对比。[182] 同样地，通过飞秒时间分辨光电子能谱技术，研究者对 2 硅氧化物/硅基底 2 上单层硒化钨的载流子动力学特性进行了深入探究，揭示了电子间谷电子-声子散射的作用机制。[183]

在单层或少层过渡金属硫属化合物（TMDs）中，非辐射发射过程中的能量耗散在载流子的超快过程中起着至关重要的作用。通常，在层状TMDs光激发后存在两种非辐射能量通道 [184]（见 **图13**）：1）通过电子-声子散射对热载流子进行超快冷却并随后形成激子，以及2）激子在表面的非辐射复合。虽然MoS的热化 2 由于电子-声子散射导致激子共振能量红移，但热载流子冷却释放的多余能量可在约2皮秒内耗散并转移至声子（路径1），而过程2）是负责从表面向外部声子耗散能量的Shockley-Read-Hall复合，耗散时间约为9皮秒。

此外，声子也被发现对激子的产生至关重要，激子是一种准粒子，对层状半导体材料的光学性质起着重要作用。通过实验测量去相位，可以得到激子-声子的强度。

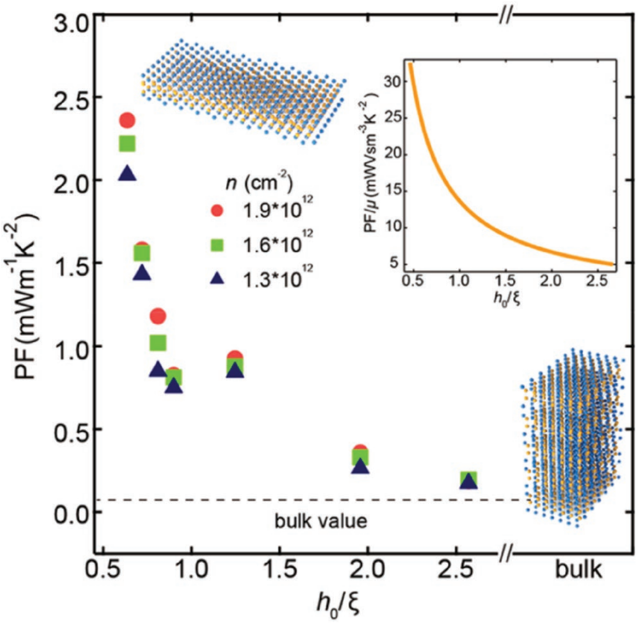

Figure 12. Power factor dependence on h_0/ξ at different carrier concentration.[176] h_0 is the quantum confinement length and ξ the de Broglie wavelength. Power factor becomes saturated to bulk value when h_0/ξ is far large than 1. When the sample thickness is much smaller than the de Broglie wavelength, there is significant increase in thermal power due to quantum confinement. Reproduced with permission.[176] Copyright 2018, American Chemical Society.

times,[185] exciton linewidths and mobility,[186] and photoluminescence.[187] Different from excitons in bulk GaAs materials where the excitons are predominantly located at the center of the Brillouin zone thus carrying zero momentum, excitons in TMDs are formed at the noncentral points in the momentum space.[188] This induces dramatic differences in the pathways of the exciton creation between GaAs and TMDs. For these nonzero wavevector excitons, the subsequent relaxation to the zero wavevector momentum involves emission of acoustic phonons or longitudinal optical (LO) phonons. A model within the framework of Fermi's Golden rule was established to derive the formation dynamics of excitons from free carriers in TMDs.[189] Depending on the density of the electrons and holes, nonzero momentum excitons could be generated from free electron–hole pairs in layered materials via the longitudinal optical phonons coupling of charge carriers at high temperature (≈120 K). On the other hand, in monolayer TMDs, excitons can be relaxed by phonons by means of the deformation potential or in the process of piezoelectric coupling. This relaxation effect could be as important as those by defect-assisted scattering or trapping effect by the surface states.[190] Photoluminescence study of monolayer MoS_2 highlighted the importance of exciton-phonon coupling.[191]

In addition to phonon–phonon, phonon-boundary and phonon-defect scattering, the phonon lifetime can be modulated by other factors as well. In particular, due to their atomic thickness, the phonon properties of 2D materials are extremely sensitive to environment, including substrate, electrical field, etc. One example is for free-standing and supported graphene. As well known, there are three acoustic phonon modes in monolayer freestanding graphene, the in-plane longitudinal acoustic mode (LA), in-plane transverse acoustic mode and out-of-plane flexural mode (ZA), and at Brillouin zone center, the energy of these acoustic modes is zero. However, for supported graphene, while the in-plane LA and TA modes are slightly influenced, the ZA mode is flattened and shifted. Obviously, a new flexural mode, called ZA' mode appears with energy of 6 THz at Γ point. This is because the substrate can break the translational invariance at the out-of-plane direction. The mismatch in flexural modes can induce significant resistance for heat flow through boundary between free-standing and supported graphene.[192] Furthermore, due to the different lattice constant between the supported 2D materials and substrate, there exists strain (stress) in the 2D materials. Usually, tensile strain results in softening of the bonds, reducing the phonon group velocity and enhancing the phonon–phonon anharmonic scattering strength. The combination of all these factors will reduce the thermal conductivity of 2D materials.[193] However, for thermal conductance of certain 2D materials, for example, borophene, tensile strain may increase its thermal conductance along certain crystal axis, because of

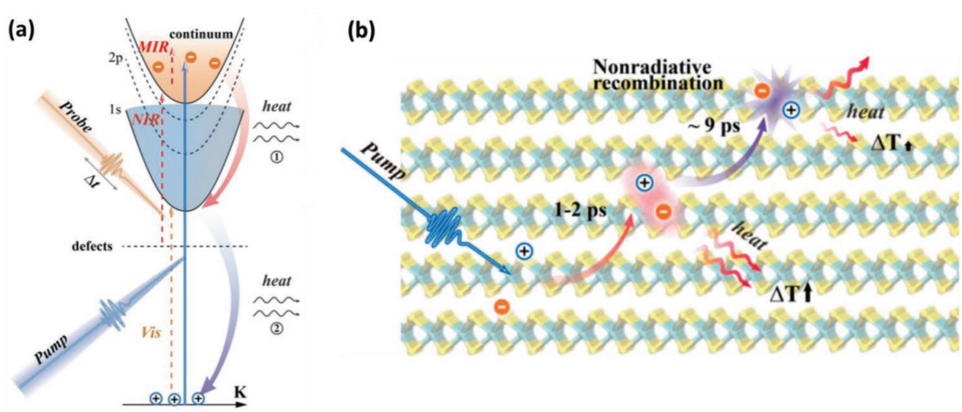

Figure 13. a) Schematic for visible (vis), near-infrared (NIR), and mid-infrared (MIR) detection irradiation in MoS_2. Possible optical transitions are marked by dashed and solid straight arrows caused by the pump pulse and probe pulse. b) Sketch of nonradiative energy channels that follow the ultrafast excitation. Reproduced with permission.[184] Copyright 2018, American Chemical Society.

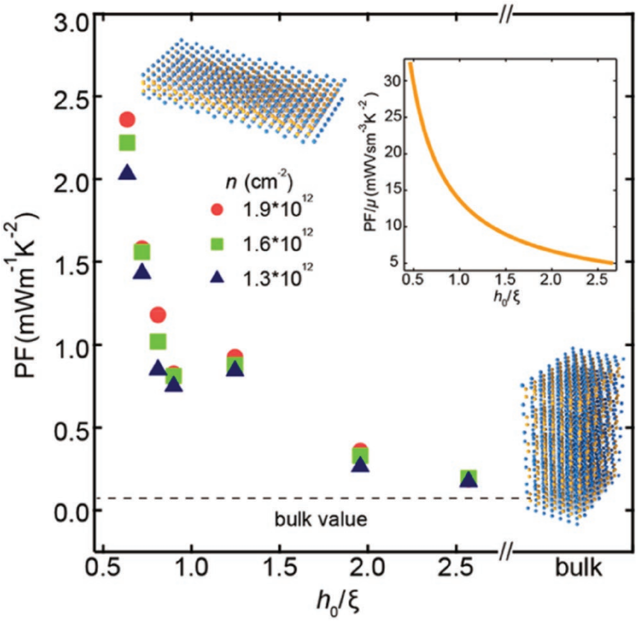

图12. 不同载流子浓度下功率因数随 h_0/ξ 的变化关系。[176] h_0 为量子限制长度，ξ 为德布罗意波长。当 h_0/ξ 远大于1时，功率因数达到体材料的饱和值。当样品厚度远小于德布罗意波长时，量子限制效应会导致热功率显著增加。经授权转载，[176] 版权所有2018年美国化学会。

在不同时间尺度下，[185] 兴子的线宽和迁移率，[186] 以及光致发光特性。[187] 与块体砷化镓材料中主要位于布里渊区中心因而具有零动量的激子不同，过渡金属硫属化合物（TMDs）中的激子形成于动量空间的非中心点。[188] 这导致砷化镓与TMDs在激子生成路径上存在显著差异。对于这些非零波矢激子，其后续弛豫至零波矢动量的过程涉及声学声子或纵向光学（LO）声子的发射。基于费米黄金法则框架建立了一个模型。

推导出过渡金属硫属化合物中自由载流子激子的形成动力学。[189] 根据电子和空穴的密度，通过高温（≈120 K）下载流子的纵向光学声子耦合，层状材料中的自由电子-空穴对可产生非零动量激子。另一方面，在单层过渡金属硫属化合物中，激子可通过形变势或压电耦合过程被声子弛豫。这种弛豫效应可能与缺陷辅助散射或表面态捕获效应同样重要。单层MoS [190] 光致发光研究 $_2$ 强调了激子-声子耦合的重要性。[191]

除了声子-声子散射、声子-边界散射和声子-缺陷散射外，声子寿命还可能受到其他因素的调控。特别是由于二维材料的原子厚度特性，其声子性质对基底、电场等环境因素极为敏感。以自由支撑石墨烯和支撑石墨烯为例，众所周知，单层自由支撑石墨烯存在三种声学模式：面内纵向声学模式（LA）、面内横向声学模式和面外弯曲模式（ZA）。在布里渊区中心，这些声学模式的能量均为零。然而对于支撑石墨烯而言，虽然面内LA和TA模式受到轻微影响，但ZA模式会出现展宽和位移现象。值得注意的是，在Γ点处会形成能量为6 THz的新弯曲模式——ZA ʻ模式。这是因为基底会破坏面外方向的平移对称性。这种弯曲模式的差异会导致自由支撑石墨烯与支撑石墨烯之间的热流边界产生显著的热阻效应。[192] 此外，由于支撑的二维材料与基底之间存在不同的晶格常数，这些材料内部会产生应变（应力）。通常情况下，拉伸应变会导致化学键软化，降低声子群速度并增强声子间非谐散射强度。这些因素共同作用会降低二维材料的热导率。[193] 然而对于某些特定二维材料（如硼烯），拉伸应变可能沿着特定晶体轴方向增强其热导率，因为

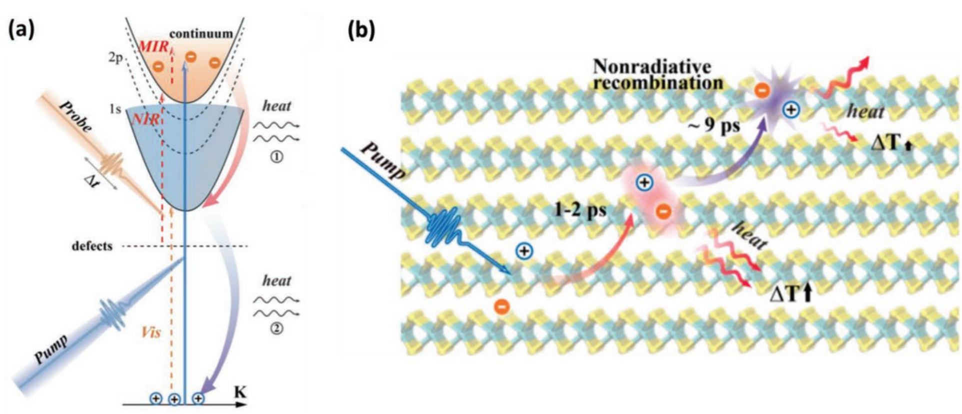

图13. a) MoS $_2$ 中可见光（vis）、近红外（NIR）和中红外（MIR）检测辐照的示意图。由泵浦脉冲和探测脉冲引起的可能光学跃迁用虚线和实线箭头标记。b) 超快激发后非辐射能量通道的示意图。经许可转载。[184] 版权所有2018年美国化学会。

the change in the bond environment[194] and for monolayer silicene, tensile strain would greatly increase its thermal conductivity due to the enhancement in the acoustic phonon lifetime.[195]

Moreover, the interactions between electron and phonon would affect the thermal transport property of 2D materials as well. In electronic devices based on 2D materials an electric field always exists. For some polar materials, the electric field may have a direct influence on their phonon vibrational modes, consequently affecting the thermal conductivity. For nonpolar materials, electric field may provide indirect effect on thermal conductivity through electron–phonon coupling.[196] Using fully first-principles calculations, Liao et al.[197] studied the effect of electron–phonon interaction on phonon transport, and compared this effect with the intrinsic phonon–phonon anharmonic interaction. Using silicon as an example, they found a significant reduction in the lattice thermal conductivity due to electron–phonon interaction when the carrier density exceeded 10^{19} cm^{-3}. It is worth mentioning that for p-type silicon with hole density of 10^{21} cm^{-3}, electron–phonon coupling can induce 45% reduction in its room temperature thermal conductivity. Considering the fact that this is a typical range of carrier concentration in the electronic devices and thermoelectric devices, the electron–phonon coupling cannot be ignored in a study of thermal conductivity of semiconductors. The impact of electron–phonon interaction on thermal conduction of 2D materials deserves future studies.

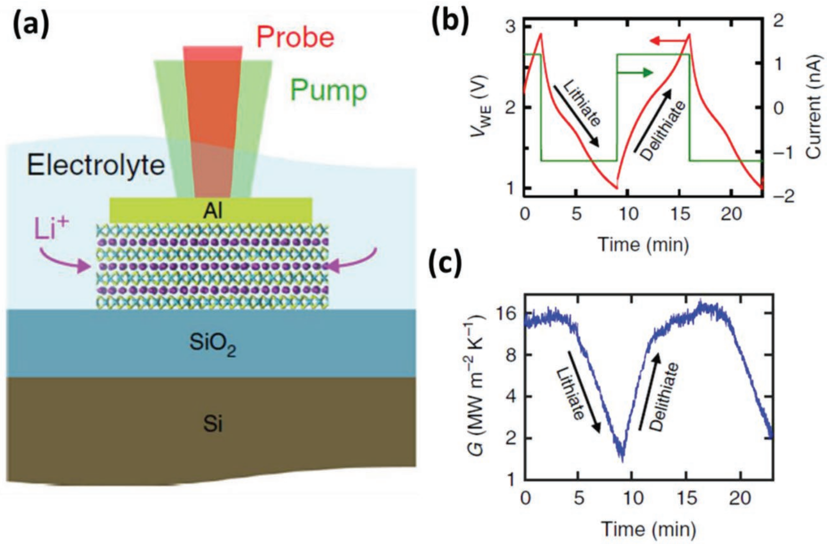

Figure 14. a) Illustration of the thermal transistor under operation. b) The MoS_2 working electrode VWE measured during the electrochemical cycle. c) Cross-plane thermal conductance during the electrochemical cycle. Reproduced with permission.[206] Copyright 2018, Nature Research.

4.3. Thermal Transistor Devices

Engineering the physical characteristics of phonons provides a powerful way to realize designable thermal functional materials. Furthermore, achieving active control of heat flow in a manner analogous to electronic circuits represents a game changer in engineering energy transport. Theoretical models to realize various types of thermal devices have been developed.[198,199] Because of the advantages of high thermal conductivity, single-atom thickness and tunable shape, graphene-based thermal rectifier[200–204] and thermal modulator[205] have been explored both theoretically and experimentally.

Recently, Sood et al. demonstrated experimentally a thermal transistor based on MoS_2 thin film actuated by reversible electrochemical intercalation of Li ions.[206] Conceptually, thermal transistor is a device whose thermal conductance can be modulated via an external stimulus, such as electrical voltage. For the MoS_2 based thermal transistor, the on-state corresponds to the pristine MoS_2 and the off-state corresponds to the Li-intercalated MoS_2. **Figure 14**a shows the schematic of the device under operation. It is a 10 nm thick few-layer MoS_2 film, and $LiPF_6$ in ethylene carbonate/diethyl carbonate serves as the liquid electrolyte. Charging/discharging the cell is controlled by electrical current in a manner of real-time modulation. As shown in Figure 14b, during the lithiation step, the MoS_2 working electrode decreases as Li ions enter the MoS_2 films. Consequently, the cross-plane thermal conductivity decreases from about 15 to 1.6 MW m^{-2} K^{-1} as shown in Figure 14c. When the current is reversed during the delithiation step, Li ions are removed from the MoS_2 film, and the cross-plane thermal conductance increases back to the pre-lithiation value. The on/off ratio is about 10 times between the lithiated and delithiated states.

The underlying mechanism was examined using first-principles calculations. The pre-lithiation MoS_2 film is in 2H phase with ABAB stacking sequence. Upon intercalation, Li ions occupy the octahedral sites inside the interlayer gap, forming 2H-Li_1MoS_2 phase and mixed with 1T-Li_1MoS_2 phase with stacking sequence AA. As shown in **Figure 15**a–c, lithiation

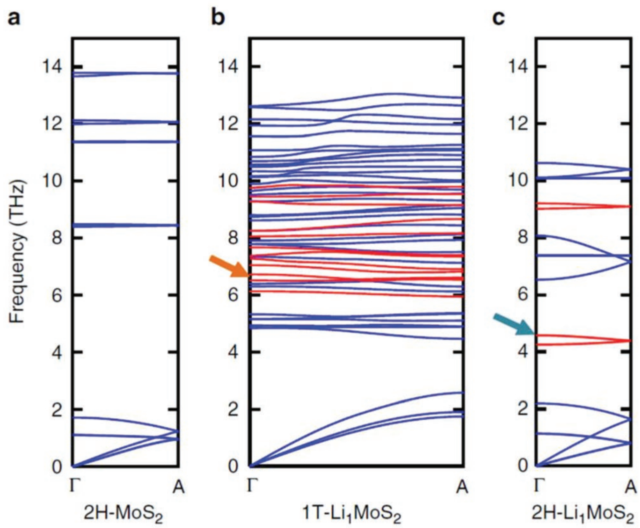

Figure 15. Phonon dispersions along the cross-plane direction. a) 2H-MoS_2, b) 1T-Li_1MoS_2, and c) 2H-Li_1MoS_2. Blue curves show the contribution from MoS_2 and red curves show that from Li atoms. Reproduced with permission.[206] Copyright 2018, Nature Research.

对于单层硅烯来说，由于声子寿命的增强，[194] 在键合环境变化时，拉伸应变会大大增加其热导率。[195]

此外，电子与声子之间的相互作用同样会影响二维材料的热传导特性。在基于二维材料的电子器件中，电场始终存在。对于某些极性材料，电场可能直接影响其声子振动模式，从而改变热导率。而对于非极性材料，电场则可能通过电子-声子耦合间接影响热导率。[196] 通过完全基于第一性原理的计算，廖等人[197]
研究了电子-声子相互作用对声子输运的影响，
并将其与声子-声子的本征相互作用进行了对比
分析。
非谐波相互作用。以硅为例，他们发现显著的
减少
当载流子浓度超过10 ^{19} cm $^{-3}$ 时，电子-声子相互作用会导致晶格热导率发生显著变化。值得注意的是，对于空穴浓度为10 ^{21} cm $^{-3}$ 的p型硅材料，电子-声子耦合作用可使其室温热导率降低45%。考虑到这一浓度范围在电子器件和热电器件中普遍存在，研究半导体热导时电子-声子耦合作用的影响不容忽视。电子-声子相互作用对二维材料热传导特性的影响，值得未来深入研究。

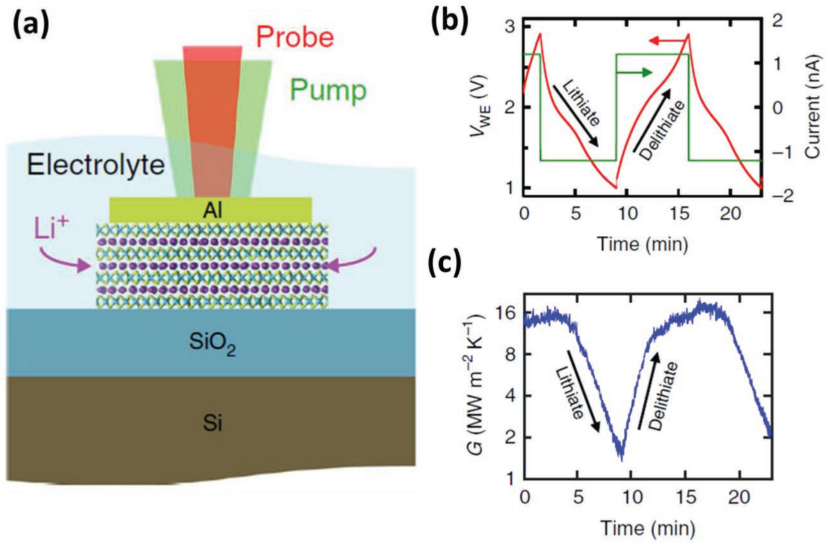

图14. a) 运行中热晶体管的示意图。b) 电化学循环期间测量的MoS $_2$ 工作电极 VWE 。c) 电化学循环期间的横截面热导率。经许可转载。[206] 版权所有2018年，Nature Research。

4.3. 热晶体管器件

通过调控声子的物理特性，为设计可定制的热功能材料提供了强大手段。更关键的是，这种类似电子电路的热流主动控制技术，堪称能源传输工程领域的颠覆性突破。目前已有多种热器件的理论模型被成功建立。[198,199] 由于石墨烯材料具备高导热性、单原子厚度和可调形貌等优势，基于石墨烯的热整流器[200–204] 和热调制器[205] 已在理论与实验层面获得深入探索。

最近，Sood等人通过实验展示了一种基于MoS $_2$ 薄膜的热晶体管，该薄膜通过Li离子的可逆电化学嵌入驱动。[206] 概念上，热晶体管是一种通过外部刺激（如电压）来调节热导率的器件。 对于基于MoS $_2$ 的热晶体管，导通状态对应原始MoS $_2$ ，关断状态对应Li嵌入的MoS $_2$ 。**图14a** 展示了器件的工作示意图。该器件采用10纳米厚的少层MoS $_2$ 薄膜，碳酸乙烯酯/碳酸二乙酯 $_6$ 中的LiPF作为液态电解质。通过实时电流调控实现电池的充放电。如图14b所示，在锂化步骤中，MoS $_2$ 工作电极

随着Li离子进入MoS $_2$ 薄膜，其导热率降低。因此，如图14c所示，面内热导率从约15降至1.6 MW m $^{-2}$ K $^{-1}$ 。当在脱锂步骤中反转电流时，Li离子从MoS $_2$ 薄膜中被移除，面内热导率回升至脱锂前的数值。锂化与脱锂状态之间的开/关比约为10倍。

通过第一性原理计算研究了其底层机制。预锂化的MoS $_2$ 薄膜处于2H相，具有 ABAB 堆叠序列。经过插层后，Li离子占据层间间隙内的八面体位点，形成2H-Li $_1$ MoS $_2$ 相，并与具有AA堆叠序列的1T-Li $_1$ MoS $_2$ 相混合。如 **图15** a–c所示，锂化过程

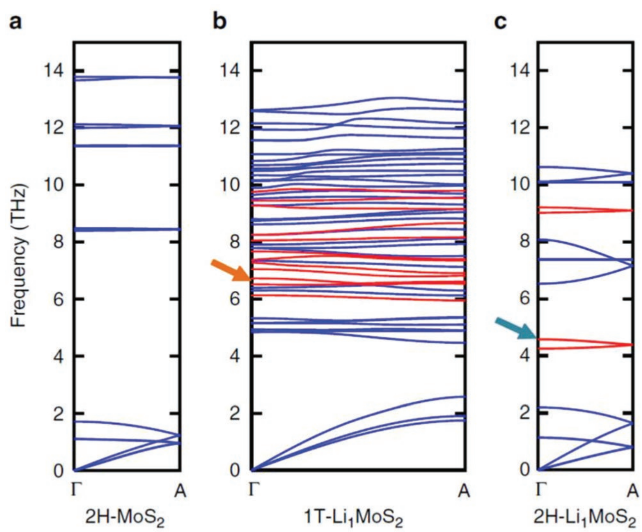

图15. 横向平面方向的声子色散。a) 2H-MoS $_2$ ，b) 1T-Li $_1$ MoS $_2$ ，c) 2H-Li $_1$ MoS $_2$ 。蓝色曲线显示MoS $_2$ 的贡献，红色曲线显示Li原子的贡献。经许可转载。[206] 版权所有2018，Nature Research。

gives rise to several flat bands, results in decreased group velocity and increased phonon scattering rates according to phonon rattling effect,[207] and consequently leads to a reduction in thermal conductivity. Moreover, lithiation may induce disordered stacking among MoS_2 film, lead to significant reduction in cross-plane thermal conductivity. Thus the observed thermal transistor is due to lithiation induced phonon scattering, reduction in group velocity and stacking disorder.

5. Conclusions and Outlook

In this article, various thermal issues related to 2D semiconductors in various electrical applications are discussed. With the rapid discovery of various 2D semiconductors, better understanding of their thermal transport properties is necessary for thermal management and energy conversion. We consider the effect of various conditions on phonon transport of channeling materials, such as strain, orientations and dimensions effect, etc. We have focused on the experimental thermal property measurement of 2D semiconductors obtained by various techniques, and the inconsistency with theoretical predication implies that new techniques with improved accuracy of measurement are needed as well as better device fabrication of 2D semiconductors. Furthermore the interfacial thermal resistance across channel/substrate and channel/contact has been discussed. Other than the direct influence on thermal management and dissipation, thermal properties related to the applications of 2D semiconductors have been discussed, including the TE effect, electron–phonon coupling effect in photoelectric phenomena, and thermal transistor devices.

Advanced fabrication techniques have made the preparation of various stable 2D materials with novel properties possible, and multistage circuit integration of 2D materials has been demonstrated recently, which is a crucial step toward 2D applications in the next-generation electronic devices. Together with the interesting electrical/optical/optoelectronic properties of novel 2D semiconductors, their thermal and thermoelectric properties should be considered as well, which is believed to be an immature field and needs more investigations. The energy dissipation at the interface should be re-considered as well when designing the nanoscale devices based on 2D materials given the large number of interfaces involved in an integrated circuit. From the fundamental study point of view, there are currently limited options to experimentally study thermal transport in 2D semiconductors and new complementary techniques should be developed to enable more systematic studies to understand their novel phonon transport behaviors.

Acknowledgements

The authors gratefully acknowledge funding from A*STAR SERC, Grant No. 1527000015.

Conflict of Interest

The authors declare no conflict of interest.

Keywords

2D semiconductors, interface thermal resistance, phonon transport, thermal functional device

Received: May 16, 2019
Revised: August 2, 2019
Published online:

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这导致了多个平带的产生，根据声子抖动效应，导致群速度降低和声子散射率增加，[207] 并最终导致热导率降低。此外，锂化可能在MoS_2薄膜中诱导出无序堆叠，导致面内热导率显著降低。因此，观察到的热晶体管是由于锂化诱导的声子散射、群速度降低和堆叠无序所致。

5. 结论和展望

本文系统探讨了二维半导体在各类电子应用中涉及的热学问题。随着新型二维半导体材料的快速涌现，深入理解其热输运特性对实现热管理与能量转换至关重要。我们重点分析了应变、取向及尺寸效应等不同条件对沟道材料声子传输的影响。通过实验测量发现，现有技术获取的二维半导体热学参数与理论预测存在显著偏差，这表明需要开发更精准的测量技术并优化器件制备工艺。此外，我们还深入研究了沟道/衬底界面及沟道/接触界面的热阻特性。除直接影响热管理与能量耗散外，文中还探讨了与二维半导体应用相关的热学特性，包括热电效应、光电现象中的电子-声子耦合效应以及热晶体管器件等关键领域。

先进的制备技术使得制备具有新型特性的多种稳定二维材料成为可能，近期还展示了二维材料的多级电路集成技术，这为下一代电子设备中的二维应用迈出了关键一步。除了新型二维半导体引人注目的电学/光学/光电子特性外，其热学和热电特性也需重点考量——这一领域仍处于起步阶段，亟待深入研究。在基于二维材料设计纳米级器件时，由于集成电路涉及大量界面接触，界面处的能量耗散问题也需重新评估。从基础研究的角度来看，目前实验研究二维半导体热传输的方法选择有限，亟需开发新的互补技术，以便更系统地探究其独特的声子传输行为。

致谢

作者衷心感谢新加坡科技研究局（A*STAR SERC）提供的资助（项目编号：1527000015）。

违背公众利益的行为

作者声明不存在利益冲突。

关键词

二维半导体、界面热阻、声子传输、热功能器件

接收日期：2019年5月16日
修订日期：2019年8月
2日，发布日期：

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